
Coarsening Bias from Variable Discretization in Causal Functionals

Xiaxian Ou¹

Razieh Nabi¹

¹Department of Biostatistics and Bioinformatics, Emory University, Atlanta, Georgia, USA

Abstract

Causal identification functionals often require integration over conditional densities of continuous variables, such as those arising in nonparametric identification theory of total and mediated causal effects in DAGs with hidden variables. Estimating these densities and evaluating the resulting integrals can be statistically and computationally demanding. A common workaround is to discretize the continuous variable and replace integrals with finite sums. Although convenient, discretization alters the population-level functional and can induce non-negligible approximation bias, even when identification is correct. Under smoothness conditions, we show that the resulting coarsening error is first order in the bin width and arises at the level of the target functional, distinct from statistical estimation error. We propose a simple debiased coarsened functional that evaluates the outcome regression at within-bin conditional means, eliminating the leading coarsening error term and yielding a second-order approximation error. We derive plug-in and one-step estimators for this debiased coarsened functional. Simulations demonstrate substantial bias reduction and near-nominal confidence interval coverage, even under coarse binning. Our results provide a simple framework for controlling the impact of variable discretization on both parameter approximation and statistical estimation.

1 INTRODUCTION

Several causal parameters admit identification formulas that require averaging an outcome regression over the distribution of mediating variables. This structure arises in mediation analysis [Pearl, 2001, VanderWeele and Vansteelandt,

2014], path-specific effects [Avin et al., 2005, Shpitser, 2013], and nonparametric identification of causal effects in hidden-variable DAGs [Tian and Pearl, 2002, Bhattacharya et al., 2022, Richardson et al., 2023, Guo et al., 2023, Guo and Nabi, 2024, Guo et al., 2025]. When the mediator is continuous or high-dimensional, as is common in biomedical applications involving biomarkers such as BMI, blood pressure, or gene expression profiles and in social science applications involving cognitive test scores, psychometric measurements, or socioeconomic indices [Salinas et al., 2021, Huang et al., 2022, Hemade and Salameh, 2025], evaluating these identification formulas requires estimating conditional mediator densities and computing integrals with respect to them. Both tasks can be statistically challenging, and Monte Carlo approximations may be computationally intensive and unstable in moderate samples.

A range of strategies have been proposed to address these challenges. One class of approaches directly models the conditional mediator density and evaluates the identifying integral numerically, using (semi/non-)parametric estimators [Ma and Zeng, 2014, Tingley et al., 2014, Devick et al., 2022, Linero, 2022, Li et al., 2023]. These methods can be computationally demanding, and typically rely on sufficiently accurate estimation of the conditional mediator densities and, in parametric implementations, require correct density specification for consistency. Sequential regression approaches avoid explicit density estimation by rewriting the identifying functional as iterated conditional expectations, thereby reducing the problem to estimation of a collection of regression functions [Liu et al., 2025, Wang et al., 2025]. However, consistency of such estimators often requires correct specification of all the nuisance regression components. Alternatively, the identifying functional can often be expressed using inverse probability weights involving treatment probabilities and mediator density ratios, which may be rewritten via Bayes' rule. Such weighting approaches can suffer from instability when treatment probabilities are small or poorly estimated [Hong, 2025, Zhou, 2022, Liu and Ghassami, 2024]. Influence-function based estimators,

including one-step and targeted minimum loss estimators [Van der Vaart, 2000, van der Laan and Rubin, 2006], provide a principled approach to estimation bias reduction and inference with flexible nuisance estimation, and can offer robustness to some forms of model misspecification. However, such estimators are often avoided in practice due to methodological complexity.

In practice, a simpler workaround is often adopted by discretizing the mediator and replacing integrals with finite sums [Chung et al., 2022, Morera et al., 2023, Piccininni et al., 2023]. This avoids estimation of a continuous conditional density and simplifies computation. However, discretization changes the population-level parameter. Even when the causal identification assumptions hold, the discretized functional is, in general, a different parameter from the original one. In many applied analyses, the bias induced by mediator discretization is either overlooked or treated as secondary relative to computational considerations, and its impact on the target parameter is rarely quantified.

Here, we study the difference between the original causal functional and its discretized analogue, which we refer to as the coarsening error (or, when viewed relative to the original target parameter, the coarsening bias). Our primary focus is not on first-order estimation bias arising from nuisance estimation, but rather on the population-level approximation error induced by mediator discretization itself. We show that, under smoothness conditions, the naive discretized functional incurs a first-order error proportional to the maximum bin width, driven by within-bin shifts in mediator means across treatment levels. We then construct a debiased coarsened functional that evaluates the outcome regression at within-bin conditional means. A Taylor expansion argument shows that this modification eliminates the leading first-order coarsening bias term, yielding a second-order remainder in the bin width. For statistical estimation, we derive plug-in and one-step estimators and establish their asymptotic properties. Simulation studies demonstrate substantial bias reduction and improved coverage relative to naive discretization, even under coarse binning. More broadly, our results provide a formal foundation for controlling the impact of discretization on both parameter approximation and statistical estimation.

2 TARGET ESTIMANDS

Let A denote a binary exposure taking values in $\{a_0, a_1\}$, C observed covariates with support $\mathcal{C}$ and marginal distribution P_C , M a mediator, and Y the outcome. Define the outcome regression $\mu(m, a, c) = \mathbb{E}(Y \mid M = m, A = a, C = c)$, the conditional mediator density $f_{M \mid A, C}(m \mid a, c) = p(M = m \mid A = a, C = c)$, and the propensity score $\pi(a \mid c) = p(A = a \mid C = c)$. Let $Q = \{\mu, f_{M \mid A, C}, \pi\}$ collect the nuisance functions. The specific causal functionals considered below may depend only on subsets of Q . For each $c \in \mathcal{C}$, define

$$\theta(Q)(c) = \int \mu(m, a_1, c) f_{M \mid A, C}(m \mid a_0, c) dm. \quad (1)$$

The conditional functional $\theta(Q)(c)$ serves as a building block for several causal parameters. In particular, define

$$\begin{aligned} \psi(Q) &= \int \theta(Q)(c) p(c) dc, \\ \gamma(Q) &= p(a_0) \mathbb{E}(Y \mid a_0) + \int \theta(Q)(c) \pi(a_1 \mid c) p(c) dc. \end{aligned} \quad (2)$$

Under standard identification assumptions (see Appendix C.1), $\psi(Q)$ corresponds to the mediation identification functional for $\mathbb{E}(Y(a_1, M(a_0)))$, while $\gamma(Q)$ corresponds to the front-door identification functional for $\mathbb{E}(Y(a_0))$, where $Y(a, m)$ denotes the potential outcome under $A = a$ and $M = m$ [Pearl, 2001, 1995, Guo et al., 2023, Guo and Nabi, 2024].

Given n i.i.d. observations $(C_i, A_i, M_i, Y_i)_{i=1}^n$ and nuisance estimates collected in $\hat{Q}$, plug-in estimators of $\psi(Q)$ and $\gamma(Q)$ are given by

$$\begin{aligned} \psi(\hat{Q}) &= \frac{1}{n} \sum_{i=1}^n \theta(\hat{Q})(C_i), \\ \gamma(\hat{Q}) &= \frac{1}{n} \sum_{i=1}^n \{I(A_i = a_0)Y_i + \theta(\hat{Q})(C_i) \hat{\pi}(a_1 \mid C_i)\}. \end{aligned} \quad (3)$$

When M is continuous, evaluation of $\theta(\hat{Q})(c)$ requires computing $\theta(\hat{Q})(c) = \int \hat{\mu}(m, a_1, c) \hat{f}_{M \mid A, C}(m \mid a_0, c) dm$, which may involve numerical integration or Monte Carlo approximation under an estimated working model for the conditional density $f_{M \mid A, C}$. In many practical settings, this can be statistically and computationally challenging. A common simplification is to coarsen the mediator by discretization.

Next, we formalize such coarsened functionals and analyze the resulting coarsening errors.

3 COARSENEST ESTIMANDS

For clarity we present the development for a univariate mediator $M \in \mathbb{R}$; extensions to multivariate mediators are discussed in Appendix B. For simplicity, we assume throughout that the support of M is contained in a compact interval $\mathcal{M} = [\ell, u] \subset \mathbb{R}$.

Let $h : \mathbb{R} \rightarrow \{1, \dots, K\}$ be a measurable discretization map that partitions the support of M into disjoint bins $\mathcal{B}_k = \{m : h(m) = k\}$, $k \in \{1, \dots, K\}$. Let $w_k = \sup_{m_1, m_2 \in \mathcal{B}_k} |m_1 - m_2|$ denote the bin width and define $\tilde{M} = h(M)$. The rule h may correspond, for instance, to thresholding M at a fixed or data-dependent cutoff, but more generally can encode any measurable partition. For example, if M is BMI and the observed values lie in the interval $[\ell, u]$, one may define $K = 4$ bins $\mathcal{B}_1 = [\ell, 18.5)$, $\mathcal{B}_2 = [18.5, 25)$, $\mathcal{B}_3 = [25, 30)$, and $\mathcal{B}_4 = [30, u]$, with

$\widetilde{M} = 1, \dots, 4$ denoting underweight, healthy weight, overweight, and obesity, respectively.

For each bin k , define $\mu_k(a, c) = \mathbb{E}(Y \mid \widetilde{M}=k, A=a, C=c)$ and $g_k(a, c) = p(\widetilde{M}=k \mid A=a, C=c)$. The coarsened analogue of (1) is

$$\theta_h(Q)(c) = \sum_{k=1}^K \mu_k(a_1, c) g_k(a_0, c). \quad (4)$$

The coarsened versions of estimands in (2), denoted by $\psi_h(Q)$ and $\gamma_h(Q)$, are obtained by replacing θ with θ_h . We primarily focus on θ since coarsening behavior propagates to ψ and γ through θ by averaging over C .

Given our notation, we can write $\theta(Q)(c)$ in (1) as

$$\sum_{k=1}^K \left\{ \int \mu(m, a_1, c) p(m \mid \widetilde{M} = k, a_0, c) dm \right\} g_k(a_0, c).$$

Let $\mu_{k,a_1}(a, c) = \mathbb{E}(\mu(M, a_1, c) \mid \widetilde{M}=k, A=a, C=c)$. Thus, $\theta(Q)(c)$ in (1) can be expressed as

$$\theta(Q)(c) = \sum_{k=1}^K \mu_{k,a_1}(a_0, c) g_k(a_0, c). \quad (5)$$

Lemma 3.1. For each $c \in \mathcal{C}$, the coarsening error defined as $\Delta_h(Q)(c) = \theta_h(Q)(c) - \theta(Q)(c)$ equals

$$\Delta_h(Q)(c) = \sum_{k=1}^K \{ \mu_k(a_1, c) - \mu_{k,a_1}(a_0, c) \} g_k(a_0, c). \quad (6)$$

If $m \mapsto \mu(m, a_1, c)$ is continuously differentiable on each $\mathcal{B}_k$ and for each fixed c there exists a finite constant $L(c) < \infty$ such that $\sup_{m \in \mathcal{B}_k} |\mu'_m(m, a_1, c)| \leq L(c)$ for all $k \in \{1, \dots, K\}$ and $L(C)$ is square-integrable under P_C , then $|\Delta_h(Q)(c)| = O(w_{\max, K})$, where $w_{\max, K} := \max_{k \in \{1, \dots, K\}} w_k$ is the maximum bin width. In particular, if M has bounded support and the K bins are chosen to have equal width, then $\Delta_h(Q)(c) = O(1/K)$.

We refer to $\Delta_h(Q)$ as the coarsening error. Since inference is ultimately aimed at $\psi(Q)$, this error may also be interpreted as a coarsening bias induced by replacing the original estimand with its coarsened analogue.

When one bin is much wider than the others, e.g., $\{0\}$ versus $(0, M_{\max}]$, the worst-case bound based on $w_{\max, K}$ can be conservative; a sharper bound uses the weighted average $\sum_{k=1}^K w_k g_k(a_0, c)$. See a proof in Appendix C.2.

To understand the magnitude of coarsening error in (6) and develop corrections for the resulting coarsening bias, we take a closer look at the difference between $\mu_k(a_1, c)$ and $\mu_{k,a_1}(a_0, c)$.

Assume μ is twice continuously differentiable on each $\mathcal{B}_k$, and define the within-bin mean

$$m_k(a, c) = \mathbb{E}(M \mid A = a, C = c, \widetilde{M} = k). \quad (7)$$

Under the conditions of Lemma 3.1, a Taylor expansion of $\mu(M, a_1, c)$ around m_k yields

$$\begin{aligned} & \mu_k(a_1, c) - \mu_{k,a_1}(a_0, c) \\ &= \mu'_m(m_k(a_1, c), a_1, c) \{m_k(a_1, c) - m_k(a_0, c)\} + O(w_k^2), \end{aligned} \quad (8)$$

where μ'_m is the derivative of μ w.r.t m ; see Appendix C.2.

Equation (8) reveals that the coarsening error in (6) scales linearly with the shift in within-bin mediator means between treatment levels and the local slope of the outcome regression in M . The higher-order terms are proportional to the curvature of μ and the within-bin spread of M ; under mild smoothness conditions, these remainder terms are of second order in the bin width. Therefore $\Delta_h(Q)(c)$ admits a first-order approximation with remainder $O(w_{\max, K}^2)$. In the following remark, we re-express the coarsening error via an exact covariance representation.

Remark 3.2. $\Delta_h(Q)(c)$ admits an exact representation in terms of the within-bin covariance between (i) the conditional outcome surface $\mu(M, a_1, c)$ and (ii) the treatment-induced density ratio $r_k(M|c) = p(M \mid \widetilde{M} = k, A = a_0, C = c) / p(M \mid \widetilde{M} = k, A = a_1, C = c)$. The coarsening error vanishes if within every bin, these two quantities are uncorrelated under the distribution of M given $(\widetilde{M} = k, A = a_1, C = c)$. This occurs, for instance, if μ is locally constant in M within bins, or if the conditional distribution of M does not differ between $A = a_0$ and $A = a_1$ inside each bin; see Appendix C.3.

The plug-in estimate of the coarsened estimand $\theta_h(Q)(c)$ is

$$\theta_h(\widehat{Q})(c) = \sum_{k=1}^K \widehat{\mu}_k(a_1, c) \widehat{g}_k(a_0, c), \quad (9)$$

where $\widehat{\mu}_k(a, c)$ and $\widehat{g}_k(a, c)$ estimate $\mu_k(a, c)$ and $g_k(a, c)$, e.g., via regressions of Y on $(\widetilde{M}, A, C)$ and $\widetilde{M}$ on (A, C) . The plug-in estimators $\psi_h(\widehat{Q})$ and $\gamma_h(\widehat{Q})$ are obtained by replacing $\theta(\widehat{Q})$ with $\theta_h(\widehat{Q})$ in (3).

We decompose $\theta_h(\widehat{Q})(c) - \theta(Q)(c)$ as:

$$\underbrace{\{\theta_h(\widehat{Q})(c) - \theta_h(Q)(c)\}}_{\text{estimation error}} + \underbrace{\{\theta_h(Q)(c) - \theta(Q)(c)\}}_{\text{coarsening error}}. \quad (10)$$

Assume that for fixed h , the estimator $\theta_h(\widehat{Q})$ satisfies

$$\|\theta_h(\widehat{Q}) - \theta_h(Q)\|_{L_2(P_C)} = O_p(n^{-1/2}),$$

where $\|g\|_{L_2(P_C)}^2 = \mathbb{E}(g(C)^2)$. Then, by Lemma 3.1,

$$\|\theta_h(\widehat{Q}) - \theta(Q)\|_{L_2(P_C)} = O_p(n^{-1/2}) + O(w_{\max, K}).$$

Hence the asymptotic bias is driven by the coarsening error $\Delta_h(Q)$, which is first order in the bin width.

Let $K = K_n$ increase with n . If M has bounded support and equal-width bins are used, then $w_{\max, K_n} = O(1/K_n)$. If

$K_n \rightarrow \infty$ and $w_{\max, K_n} = o(n^{-1/2})$, the coarsening error is negligible relative to sampling error. Under equal-width binning, this requires $K_n/\sqrt{n} \rightarrow \infty$.

4 DEBIASED COARSENEDED ESTIMANDS

We introduce a simple correction that removes the leading coarsening error by evaluating the outcome regression $\mu(\cdot, a_1, c)$ at within-bin representatives of the mediator under $A = a_0$. Given $m_k(a, c)$ in (7), define the debiased coarsened functional

$$\tilde{\theta}_h(Q)(c) = \sum_{k=1}^K \mu(m_k(a_0, c), a_1, c) g_k(a_0, c). \quad (11)$$

Taking a first-order Taylor expansion of $\mu(M, a_1, c)$ around $m_k(a_0, c)$ and then taking expectations conditional on $(\tilde{M} = k, A = a_0, C = c)$ motivates the approximation of $\mu_{k, a_1}(a_0, c)$ by $\mu(m_k(a_0, c), a_1, c)$; see Appendix C.4. Thus (11) replaces the in-bin conditional expectation of $\mu(M, a_1, c)$ under $A = a_0$ by its evaluation at the within-bin mean under the same distribution.

The debiased coarsened causal parameters $\tilde{\psi}_h(Q)$ and $\tilde{\gamma}_h(Q)$ are obtained from (2) by replacing θ with $\tilde{\theta}_h$.

Lemma 4.1. *For each $c \in \mathcal{C}$, the coarsening error defined as $\tilde{\Delta}_h(Q)(c) = \tilde{\theta}_h(Q)(c) - \theta(Q)(c)$ equals*

$$\tilde{\Delta}_h(Q)(c) = \sum_{k=1}^K \{\mu(m_k(a_0, c), a_1, c) - \mu_{k, a_1}(a_0, c)\} g_k(a_0, c). \quad (12)$$

If $m \mapsto \mu(m, a_1, c)$ is twice continuously differentiable and $R_k(c) = \mu(m_k(a_0, c), a_1, c) - \mu_{k, a_1}(a_0, c)$, then

$$|R_k(c)| \leq \frac{1}{2} \sup_{m \in \mathcal{B}_k} |\mu''_m(m, a_1, c)| \mathbb{E}((M - m_k(a_0, c))^2 | a_0, c, k). \quad (13)$$

If $\sup_{m \in \mathcal{B}_k} |\mu''_m(m, a_1, c)| \leq L(c)$, where $L(C)$ is square-integrable under P_C , then $\tilde{\Delta}_h(Q)(c) = O(w_{\max, K}^2)$. If M has bounded support and the K bins are chosen to have equal width, then $\tilde{\Delta}_h(Q)(c) = O(1/K^2)$.

See Appendix C.4 for a proof.

Lemma 4.1 shows that the leading first-order component of the coarsening error in Lemma 3.1 is eliminated. The remaining error is proportional to the curvature of $\mu(m, a_1, c)$ and the within-bin variance of M , and is thus second order in the bin width.

The plug-in estimate of the debiased coarsened estimand $\tilde{\theta}_h(Q)(c)$ in (11) is

$$\tilde{\hat{\theta}}_h(\hat{Q})(c) = \sum_{k=1}^K \hat{\mu}(\hat{m}_k(a_0, c), a_1, c) \hat{g}_k(a_0, c), \quad (14)$$

where $\hat{m}_k(a_0, c) = \hat{\mathbb{E}}(M | A = a_0, C = c, \tilde{M} = k)$. The estimators $\tilde{\psi}_h(\hat{Q})$ and $\tilde{\gamma}_h(\hat{Q})$ are obtained by replacing $\theta(\hat{Q})$ with $\tilde{\theta}_h(\hat{Q})$ in (3).

We decompose $\tilde{\theta}_h(\hat{Q})(c) - \theta(Q)(c)$ as

$$\underbrace{\{\tilde{\theta}_h(\hat{Q})(c) - \tilde{\theta}_h(Q)(c)\}}_{\text{estimation error}} + \underbrace{\{\tilde{\theta}_h(Q)(c) - \theta(Q)(c)\}}_{\text{coarsening error}}. \quad (15)$$

Assume that for fixed h ,

$$\|\tilde{\theta}_h(\hat{Q}) - \tilde{\theta}_h(Q)\|_{L_2(P_C)} = O_p(n^{-1/2}).$$

Then, by Lemma 4.1,

$$\|\tilde{\theta}_h(\hat{Q}) - \theta(Q)\|_{L_2(P_C)} = O_p(n^{-1/2}) + O(w_{\max, K}^2).$$

For fixed K , the asymptotic bias is driven by the coarsening error $\tilde{\Delta}_h(Q)$, which is second order in the bin width.

Let $K = K_n$ increase with n . If M has bounded support and equal-width bins are used, then $w_{\max, K_n} = O(1/K_n)$, and hence $\tilde{\Delta}_h(Q)(c) = O(1/K_n^2)$. If $K_n \rightarrow \infty$ and $w_{\max, K_n}^2 = o(n^{-1/2})$, then the coarsening error is negligible relative to sampling error. Under equal-width binning, this requires $K_n/n^{1/4} \rightarrow \infty$.

Compared to the naive coarsened functional $\theta_h(Q)(c)$, which incurs first-order coarsening error of order $O(w_{\max, K})$, the debiased coarsened functional $\tilde{\theta}_h(Q)(c)$ achieves a second-order approximation of order $O(w_{\max, K}^2)$ under the stated smoothness conditions. Under equal-width binning, this corresponds to $O(1/K)$ versus $O(1/K^2)$.

5 SMOOTHED DEBIASED COARSENEDED ESTIMANDS

Sections 3 and 4 analyzed the approximation error induced by coarsening. We now turn to statistical estimation.

Throughout the previous sections we assumed that, for fixed h , the estimation error of the plug-in estimator is $O_p(n^{-1/2})$ in $L_2(P_C)$. Such behavior typically requires sufficiently fast convergence of all nuisance estimators. When flexible machine learning methods are used to estimate μ , π , and g_k , plug-in estimators can exhibit non-negligible first-order estimation bias.

A von Mises expansion shows that for a smooth target parameter $\psi(Q)$, $\psi(\hat{Q}) - \psi(Q) = -P\{\phi(\hat{Q})\} + R_2(\hat{Q}, Q)$, where $\phi(Q)$ is the canonical gradient, efficient influence function (EIF), and R_2 is a second-order remainder [Van der Vaart, 2000]. The leading term $-P\{\phi(\hat{Q})\}$ represents first-order estimation bias and depends directly on the quality of nuisance estimation. Achieving asymptotic linearity therefore requires sufficiently fast nuisance convergence.

Influence-function-based estimators, such as one-step and targeted minimum loss estimators, remove the first-order

bias term and attain asymptotic linearity under weaker conditions. However, to construct such estimators, the target parameter must be pathwise differentiable under the statistical model. The debiased coarsened functional $\tilde{\theta}_h(Q)(c)$ fails this requirement when M is continuous, because it involves point evaluation of $\mu(m, a, c)$ at $m_k(a_0, c)$. Point evaluation is not a continuous linear functional on $L_2(P)$ unless M is discrete, and therefore $\tilde{\psi}_h$ and $\tilde{\gamma}_h$ are not pathwise differentiable under the statistical model.

5.1 SMOOTHED ESTIMANDS AND PLUG-INS

To restore pathwise differentiability and enable influence-function-based estimation, we introduce a smoothed version of the debiased coarsened functional. We regularize the point evaluation by replacing it with a localized average. Let $\mathcal{K}$ be a symmetric kernel and let $b > 0$ be a bandwidth. Define $\mathcal{K}_b(u) = b^{-1}\mathcal{K}(u/b)$ and the normalized weight

$$\omega_{b,k}(m | a_1, c) = \frac{\mathcal{K}_b(m - m_k(a_0, c))}{\mathbb{E}(\mathcal{K}_b(M - m_k(a_0, c)) | A = a_1, C = c)}.$$

Define the localized mean

$$\mu_{b,k}(a_1, c) = \mathbb{E}(\mu(M, a_1, c) \omega_{b,k}(M | a_1, c) | A = a_1, C = c),$$

and the smoothed debiased coarsened estimand

$$\tilde{\theta}_{h,b}(Q)(c) = \sum_{k=1}^K \mu_{b,k}(a_1, c) g_k(a_0, c). \quad (16)$$

As $b \rightarrow 0$, the kernel $\mathcal{K}_b$ concentrates at $m_k(a_0, c)$ and $\tilde{\theta}_{h,b}(Q)(c) \rightarrow \tilde{\theta}_h(Q)(c)$.

The smoothed debiased coarsened parameters $\tilde{\psi}_{h,b}(Q)$ and $\tilde{\gamma}_{h,b}(Q)$ are obtained from (2) by replacing θ with $\tilde{\theta}_{h,b}$.

Lemma 5.1. *For each $c \in \mathcal{C}$, the smoothing error $\tilde{\Delta}_{h,b}^s(Q)(c) = \tilde{\theta}_{h,b}(Q)(c) - \tilde{\theta}_h(Q)(c)$ equals*

$$\tilde{\Delta}_{h,b}^s(Q)(c) = \sum_{k=1}^K \{\mu_{b,k}(a_1, c) - \mu(m_k(a_0, c), a_1, c)\} g_k(a_0, c). \quad (17)$$

Suppose: (i) $m \mapsto \mu(m, a_1, c)$ is twice continuously differentiable near $m_k(a_0, c)$ with $\sup_{m,c} |\mu''_m(m, a_1, c)| < \infty$; (ii) $\mathcal{K}$ is symmetric with $\int u \mathcal{K}(u) du = 0$ and $\int u^2 \mathcal{K}(u) du < \infty$; and (iii) $f_{M|A,C}(\cdot | a_1, c)$ is continuous and bounded away from zero near $m_k(a_0, c)$. Then, as $b \rightarrow 0$, $\tilde{\Delta}_{h,b}^s(Q)(c) = O(b^2)$. Combining with Lemma 4.1, the total approximation error $\tilde{\Delta}_{h,b}(Q)(c) = \tilde{\theta}_{h,b}(Q)(c) - \theta(Q)(c)$ satisfies $\tilde{\Delta}_{h,b}(Q)(c) = O(w_{\max,K}^2 + b^2)$.

See a proof in Appendix C.5.

The plug-in estimator for $\tilde{\theta}_{h,b}(Q)(c)$ is

$$\tilde{\theta}_{h,b}(\hat{Q})(c) = \sum_{k=1}^K \hat{\mu}_{b,k}(a_1, c) \hat{g}_k(a_0, c), \quad (18)$$

where $\hat{\mu}_{b,k}(a_1, c) = \hat{\mathbb{E}}(\hat{\mu}(M, a_1, c) \hat{\omega}_{b,k}(M | a_1, c) | a_1, c)$.

We can decompose $\tilde{\theta}_{h,b}(\hat{Q})(c) - \theta(Q)(c)$ as

$$\underbrace{\{\tilde{\theta}_{h,b}(\hat{Q})(c) - \tilde{\theta}_{h,b}(Q)(c)\}}_{\text{estimation error}} + \underbrace{\{\tilde{\theta}_{h,b}(Q)(c) - \theta(Q)(c)\}}_{\text{total approximation error}}. \quad (19)$$

Assume that for fixed h and b ,

$$\|\tilde{\theta}_{h,b}(\hat{Q}) - \tilde{\theta}_{h,b}(Q)\|_{L_2(P_C)} = O_p(n^{-1/2}).$$

Then, by Lemma 5.1,

$$\|\tilde{\theta}_{h,b}(\hat{Q}) - \theta(Q)\|_{L_2(P_C)} = O_p(n^{-1/2}) + O(w_{\max,K}^2 + b^2).$$

Thus, for fixed K and b , the asymptotic bias is driven by $\tilde{\Delta}_{h,b}(Q)(c)$, which is second order in the bin width and quadratic in the smoothing bandwidth.

Let $K = K_n$ and $b = b_n$ depend on n . Under equal-width binning, $w_{\max,K_n} = O(1/K_n)$, so

$$\tilde{\Delta}_{h,b_n}(Q)(c) = \tilde{\theta}_{h,b_n}(Q)(c) - \theta(Q)(c) = O(1/K_n^2 + b_n^2).$$

If $K_n \rightarrow \infty$, $b_n \rightarrow 0$, and $1/K_n^2 + b_n^2 = o(n^{-1/2})$, for example $K_n = o(n^{1/4})$ and $b_n = o(n^{1/4})$, then $\tilde{\theta}_{h,b_n}(\hat{Q})$ is asymptotically equivalent to $\theta(Q)$ in $L_2(P_C)$.

5.2 INFLUENCE FUNCTION BASED ESTIMATION

As an example, consider the smoothed mediation functional

$$\tilde{\psi}_{h,b}(Q) = \int \tilde{\theta}_{h,b}(Q)(c) p(c) dc.$$

To present the influence function in a transparent form, we first treat the bin centers $m_k(a_0, c)$ as fixed quantities (e.g., midpoints or pre-specified representatives).

Theorem 5.2. *The nonparametric efficient influence function for $\tilde{\psi}_{h,b}(Q)$ when $m_k(a_0, k)$ is fixed, is*

$$\begin{aligned} \tilde{\phi}_{h,b}^{\text{fixed}}(Q)(O) &= \sum_{k=1}^K \frac{\mathbb{I}(A = a_1)}{\pi(a_1 | C)} g_k(a_0, C) \{Y \omega_{b,k}(M | a_1, C) - \mu_{b,k}(a_1, C)\} \\ &\quad + \sum_{k=1}^K \frac{\mathbb{I}(A = a_0)}{\pi(a_0 | C)} \mu_{b,k}(a_1, C) \{\mathbb{I}(\tilde{M} = k) - g_k(a_0, C)\} \\ &\quad + \{\tilde{\theta}_{h,b}(Q)(C) - \tilde{\psi}_{h,b}(Q)\}. \end{aligned}$$

See a proof in Appendix C.6.

Let $\hat{Q}$ collect all nuisance estimates. The one-step estimator of $\tilde{\psi}_{h,b}(Q)$, when $m_k(a_0, k)$ is pre-specified, is

$$\tilde{\psi}_{h,b}^{\text{one-step}}(\hat{Q}) = \tilde{\psi}_{h,b}^{\text{fixed}}(\hat{Q}) + \frac{1}{n} \sum_{i=1}^n \tilde{\phi}_{h,b}^{\text{fixed}}(\hat{Q})(O_i),$$

where $\tilde{\phi}_{h,b}^{\text{fixed}}(\hat{Q})(O_i)$ is the evaluation of the influence function at $\hat{Q}$ and O_i , and

$$\tilde{\psi}_{h,b}^{\text{fixed}}(\hat{Q}) = \frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K \tilde{\mu}_{b,k}^{\text{fixed}}(a_1, C_i) \hat{g}_k(a_0, C_i),$$

$$\tilde{\mu}_{b,k}^{\text{fixed}}(a_1, c) = \mathbb{E}[\tilde{\mu}(M, a_1, c) \omega_{b,k}(M | a_1, c) | A=a_1, C=c].$$

Under sufficient regularity conditions, $n^{-1/4}$ rates of the nuisance estimators, and cross-fitting, the one-step estimator is asymptotically linear with asymptotic variance of $\mathbb{E}(\tilde{\phi}_{h,b}^{\text{fixed}}(Q)(O)^2)$.

When $m_k(a_0, c)$ is estimated, an additional influence function contribution arises, denoted by $\tilde{\phi}_{h,b}^{\omega}(Q)(O)$:

$$\tilde{\phi}_{h,b}^{\omega}(Q)(O) = \sum_{k=1}^K \alpha_k(C) \frac{\mathbb{I}(A = a_0, \tilde{M} = k)}{\pi(a_0 | C)} \{M - m_k(a_0, C)\},$$

where $\alpha_k(c) := \frac{\partial}{\partial m} \mu_{b,k}(a_1, c)$. The full EIF for $\tilde{\psi}_{h,b}(Q)$ is therefore $\tilde{\phi}_{h,b}(Q)(O) = \tilde{\phi}_{h,b}^{\text{fixed}}(Q)(O) + \tilde{\phi}_{h,b}^{\omega}(Q)(O)$. See a proof in Appendix C.6.

6 SIMULATION STUDIES

The R code is available on GitHub.

We study estimation of the mediation functional $\psi(Q)$ in (2), with $a_1 = 1$ and $a_0 = 0$. Our goals are threefold: (i) quantify the population-level coarsening error of the discretized estimand $\theta_h(Q)(c)$ and verify the first-order scaling in Lemma 3.1; (ii) evaluate the finite-sample performance of the plug-in and the one-step estimators for the coarsened estimand $\psi_h(Q)$, and the debiased coarsened estimand $\tilde{\psi}_h(Q)$, under equal-frequency discretization; (iii) examine the sensitivity of the plug-in estimators to different discretization schemes including equal-width and predefined-width discretizations.

All simulations are conducted under the following data-generating process (DGP). The baseline covariate C takes values in $\{-2, -1, 0, 1, 2\}$ with probability mass $(0.15, 0.20, 0.18, 0.30, 0.17)$ respectively;

$$A | C \sim \text{Bernoulli}(\text{expit}(0.5C)),$$

$$M | A, C \sim N(-0.6C + 2A + 0.5AC, 1),$$

$$Y | M, A, C \sim N(\mu(M, A, C), 1),$$

where $\mu(M, A, C) = 0.8C + 1.5A + 0.75AM + 0.20MC^2 + 0.1M^3 + 0.55AC$, nonlinear in M , ensuring nonzero curvature and hence non-negligible second-order terms in the Taylor expansions underlying Lemmas 3.1 and 4.1.

Table 1: Coarsening error $\Delta_h(Q)(c)$ for equal-frequency discretizations with $K = 2$ and $K = 6$.

	C = -2	C = -1	C = 0	C = 1	C = 2
K = 2	1.103	0.871	0.876	1.522	3.390
K = 6	0.195	0.149	0.216	0.550	1.562

Simulation #1: Population-level coarsening error.

We first isolate the coarsening error characterized in Lemma 3.1, $\Delta_h(Q)(c) = \theta_h(Q)(c) - \theta(Q)(c)$. To approximate the theoretical quantities appearing in $\Delta_h(Q)(c)$, we generate a Monte Carlo sample of size $n=10^6$. No nuisance estimation is involved in this experiment. Instead, we exploit the known parametric forms to compute: $g_k(a, c)$, $m_k(a, c)$, $\mu_k(a, c)$. We consider equal-frequency discretizations of M with $K=2$ and $K=6$ bins. For $K=2$, the bins are $\mathcal{B}_1 = \{M \leq 1.21\}$ and $\mathcal{B}_2 = \{M > 1.21\}$. For $K=6$, we define: $\mathcal{B}_1 = \{M \leq -0.38\}$, $\mathcal{B}_2 = (-0.38, 0.53]$, $\mathcal{B}_3 = (0.53, 1.21]$, $\mathcal{B}_4 = (1.21, 1.81]$, $\mathcal{B}_5 = (1.81, 2.49]$, and $\mathcal{B}_6 = \{M > 2.49\}$.

Table 1 reports $\Delta_h(Q)(c)$ across levels of C . For all levels, the coarsening error decreases substantially as K increases. However, even at $K = 6$, the error remains non-negligible, illustrating that discretization can alter the target parameter even with a relatively fine partition.

Equation (8) shows that the leading term of $\Delta_h(Q)(c)$ is proportional to $\mu'_m(m_k(a_1, c), a_1, c) \{m_k(a_1, c) - m_k(a_0, c)\}$. Figure 1 plots the within-bin shifts $m_k(a_1, c) - m_k(a_0, c)$ for discretizations with $K = 2$ and $K = 6$. Increasing K reduces these shifts across bins and covariate strata, thereby attenuating the dominant first-order component of the coarsening error. The empirical behavior aligns closely with the Taylor expansion derived in Lemma 3.1.

Remark 3.2 expresses the coarsening error as a within-bin covariance between $\mu(M, a_1, c)$ and the ratio $r_k(M | c)$. Figure 2 illustrates this at $C = 0$. With finer discretization, the covariance magnitude drops substantially, giving a complementary view of the reduced coarsening error.

Together, these results empirically validate the $O(1/K)$ first-order behavior stated in Lemma 3.1.

Simulation #2: Finite-sample performance.

We now study statistical estimation via simulations with $n \in \{500, 5,000, 50,000\}$, using 1,000 Monte Carlo replications for each n . We consider equal-frequency discretizations of the mediator over a broad range of bin counts K .

For each discretization level K , we compare the plug-in and one-step estimates of the coarsened estimand $\psi_h(Q)$ and the debiased coarsened estimand $\tilde{\psi}_h(Q)$. All nuisance functions are estimated using correctly specified generalized linear models. The within-bin means $m_k(a_0, c)$ and

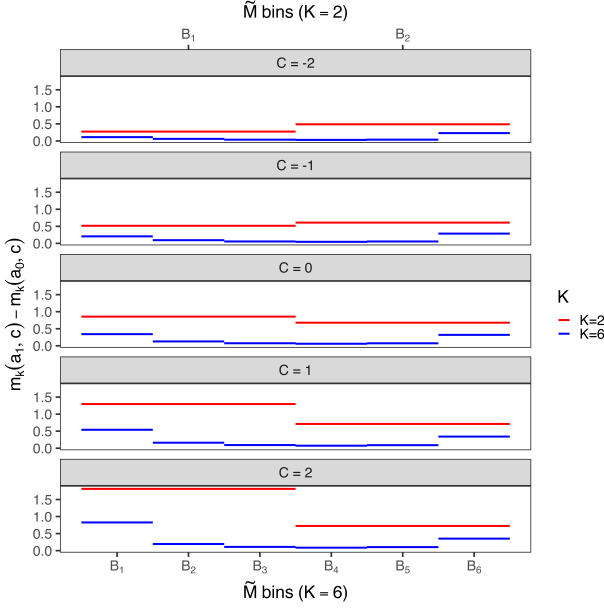


Figure 1: Within-bin differences $m_k(a_1, c) - m_k(a_0, c)$ under $K = 2$ (red, upper axis) and $K = 6$ (blue, lower axis) discretizations. Each horizontal segment represents one mediator bin; panels correspond to values of C .

bin probabilities $g_k(a, c)$ are computed under the truncated normal distribution implied by the DGP.

Figure 3 reports the results. Two distinct phenomena are evident: (i) increasing n reduces Monte Carlo variability, and (ii) for fixed K , the estimation bias of $\psi_h(\hat{Q})$ does not vanish as n increases. The persistence of bias reflects population coarsening error, not estimation error. In contrast, $\tilde{\psi}_h(\hat{Q})$ exhibits dramatically reduced bias across all sample sizes, consistent with the second-order scaling $\tilde{\Delta}_h(Q)(c) = O(w_{\max, K}^2)$ from Lemma 4.1.

Because bias patterns are qualitatively similar across sample sizes, we focus on $n = 5,000$ and report analogous results for $n = 500$ in Appendix D.2. Figure 4 summarizes absolute bias, variance, mean squared error (MSE), and coverage of 95% confidence intervals (CIs) as functions of K .

The coarsened plug-in estimator, $\psi_h(\hat{Q})$, exhibits substantial bias and poor coverage for small K , improving gradually as K increases, reflecting the $O(w_{\max, K})$ decay of the coarsening error. In contrast, the debiased coarsened estimator, $\tilde{\psi}_h(\hat{Q})$, achieves near-zero bias even for relatively small K , smaller MSE, and coverage close to the nominal level across a broad range of discretization levels. Variance is only modestly affected by the debiasing step. These findings demonstrate that evaluating μ at the within-bin mean $m_k(a_0, c)$ effectively removes the leading coarsening error without inducing meaningful variance inflation.

One-step estimators are often used to reduce estimation bias

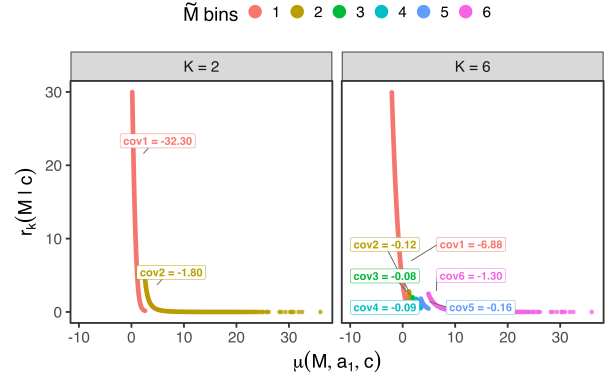


Figure 2: Within-bin covariance between $\mu(M, a_1, C)$ and $r_k(M | C)$ at $C = 0$, as formalized in Remark 3.2. Finer discretization reduces the magnitude of this covariance.

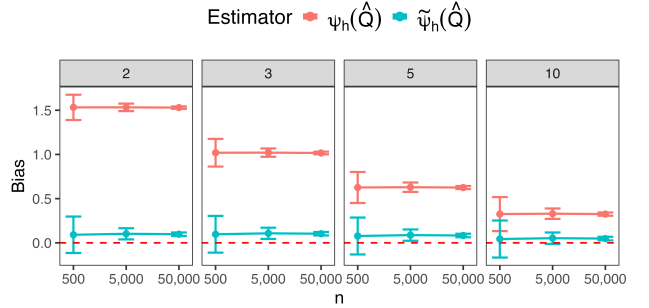


Figure 3: Bias of the plug-in estimates of the coarsened estimand, $\psi_h(\hat{Q})$, and the debiased coarsened estimand, $\tilde{\psi}_h(\hat{Q})$, as functions of sample size for several discretization levels K . Points denote the Monte Carlo mean error (bias) across 1,000 simulation replicates. Error bars show the Monte Carlo standard deviation of the estimation error.

by incorporating influence-function corrections. Define the generalized propensity score $g(a|m, c) = p(A = a | M = m, C = c)$. The one-step estimators for $\psi(Q)$ and $\psi_h(Q)$, denoted $\psi^+(\hat{Q})$ and $\psi_h^+(\hat{Q})$ are

$$\psi^+(\hat{Q}) = \frac{1}{n} \sum_{i=1}^n \left[\frac{\mathbb{I}(A_i = a_1)}{\hat{\pi}(a_0 | C_i)} \frac{\hat{g}(a_0 | M_i, C_i)}{\hat{g}(a_1 | M_i, C_i)} (Y_i - \hat{\mu}(M_i, a_1, C_i)) + \frac{\mathbb{I}(A_i = a_0)}{\hat{\pi}(a_0 | C_i)} (\hat{\mu}(M_i, a_1, C_i) - \theta(\hat{Q})(C_i)) + \theta(\hat{Q})(C_i) \right], \quad (20)$$

$$\psi_h^+(\hat{Q}) = \frac{1}{n} \sum_{i=1}^n \left[\frac{\mathbb{I}(A_i = a_1)}{\hat{\pi}(a_1 | C_i)} \frac{\hat{g}_k(a_0, C_i)}{\hat{g}_k(a_1, C_i)} (Y_i - \hat{\mu}_k(a_1, C_i)) + \frac{\mathbb{I}(A_i = a_0)}{\hat{\pi}(a_0 | C_i)} (\hat{\mu}_k(a_1, C_i) - \theta_h(\hat{Q})(C_i)) + \theta_h(\hat{Q})(C_i) \right]. \quad (21)$$

The estimator in (21) is consistent for $\psi_h(Q)$ if either (i) $\hat{\pi}$ and $\hat{g}_k$, (ii) $\hat{\pi}$ and $\hat{\mu}_k$, or (iii) $\hat{g}_k$ and $\hat{\mu}_k$ are consistent.

Note that $\psi_h^+(\hat{Q})$ still targets the coarsened functional $\psi_h(Q)$ rather than the original parameter $\psi(Q)$. To remedy this limitation, we compare: (i) $\psi_h^+(\hat{Q})$, the one-step estima-

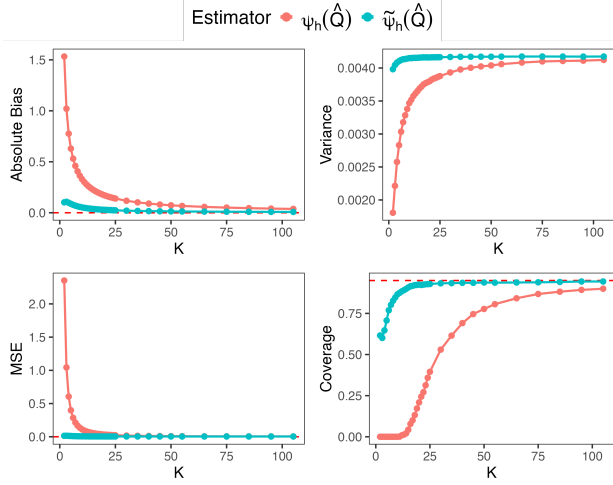


Figure 4: The performance of the coarsened plug-in estimator $\psi_h(\hat{Q})$ and the debiased estimator $\tilde{\psi}_h(\hat{Q})$ as functions of the number of mediator bins K for sample size $n = 5,000$.

tor for $\psi_h(Q)$, (ii) $\tilde{\psi}_{h_1}^+(\hat{Q})$, a naive derivative-based correction that replaces $\hat{\mu}_k$ by $\hat{\mu}(m_k(a_0, c), a_1, c)$ and $\theta_h(\hat{Q})(c)$ with $\tilde{\theta}_h(\hat{Q})(c)$ in (21), (iii) $\tilde{\psi}_{h_2}^+(\hat{Q})$, the one-step estimator that replaces $\theta(\hat{Q})(c)$ with $\tilde{\theta}_h(\hat{Q})(c)$ in (20), (iv) the coarsened plug-in estimator $\psi_h(\hat{Q})$, (v) the debiased plug-in estimator $\tilde{\psi}_h(\hat{Q})$.

Figure 5 presents results under correct specification and a range of misspecification for nuisance models. Details of the misspecification configurations are given in Appendix D.1.

The one-step estimator $\psi_h^+(\hat{Q})$ yields modest improvements in bias and coverage relative to $\psi_h(\hat{Q})$, but does not eliminate discretization bias when K is small, since the target remains $\psi_h(Q)$. The first derivative estimator $\tilde{\psi}_{h_1}^+(\hat{Q})$ behaves similarly to $\psi_h^+(\hat{Q})$, with improved coverage, but does not meaningfully reduce discretization bias in the way that $\tilde{\psi}_h(\hat{Q})$ does. In contrast, $\tilde{\psi}_{h_2}^+(\hat{Q})$ achieves uniformly low bias across different K and near-nominal coverage across scenarios except when all nuisance functions are misspecified. This strong performance arises because $\tilde{\psi}_{h_2}^+(\hat{Q})$ combines two bias-reduction mechanisms: its one-step correction targets the original parameter $\psi(Q)$, reducing first-order estimation bias due to nuisance misspecification, and the debiased coarsened functional $\tilde{\theta}(Q)(c)$ removes the leading discretization bias. Notably, although the debiased plug-in estimator can still be affected by nuisance-model misspecification, under correct specification $\tilde{\psi}_h(\hat{Q})$ performs closest to $\tilde{\psi}_{h_2}^+(\hat{Q})$ among the other four estimators, with only minor differences in bias and coverage.

These findings demonstrate that the coarsening error arises at the level of the target functional and therefore cannot be eliminated by influence-function correction alone. Correct-

ing the functional through the proposed debiased coarsened construction is essential for valid inference. The debiased plug-in estimator achieves substantial bias reduction while retaining computational simplicity and strong empirical performance when nuisance models are well estimated, which can be facilitated by flexible machine learning methods.

Simulation #3: Comparisons of discretization schemes.

Using the same DGP, we repeated the simulation procedure with 1,000 Monte Carlo replications for $n \in \{500, 5,000, 50,000\}$ and considered two alternatives to the equal-frequency discretization used above. First, we used empirical equal-width discretization, where the cut points are computed from the observed range of M within each simulated dataset. Second, we used predefined-width discretization, in which the interval $[-1.5, 3.3]$ is partitioned into K subintervals of equal length and the two boundary cut points are set to observed minimum and maximum values. For example, when $K = 3$, the resulting bins are ≤ 0.1 , $(0.1, 1.7]$, and > 1.7 . Thus, for a fixed K , all Monte Carlo replications use the same cut points.

Figure S4 displays bias trends as n increases for specific K , and Figure S5 summarizes absolute bias, variance, MSE, and coverage as functions of K for $n = 5,000$. The behavior of the coarsened estimator $\psi_h(\hat{Q})$ depends meaningfully on the discretization rule. Under empirical equal-width discretization, the bias of $\psi_h(\hat{Q})$ can increase with n for some fixed values of K , particularly when K is small, except for $K=2$. This occurs because the mediator has unbounded support, and M is unimodal. When the sample size increases, the empirical range tends to expand, leading to wider equal-width bins. In particular, wider central bins can contain a larger proportion of observations and amplify the within-bin differences in conditional mediator means, $m_k(a_1, c) - m_k(a_0, c)$. In contrast, under predefined-width discretization, the interior bin widths are fixed for each K . Consequently, the bias of $\psi_h(\hat{Q})$ remains stable as n increases for a fixed K , consistent with the behavior observed under equal-frequency discretization. However, as shown in Figure S5b, although the bias decreases as K increases, it does not converge to zero. This residual bias arises because the two tail bins continue to contribute approximation error even as the interior discretization becomes finer; in particular, these tail bins always contain the regions ≤ -1.5 and ≥ 3.3 . Across both schemes, the proposed debiased estimator $\tilde{\psi}_h(\hat{Q})$ substantially reduces bias and remains close to the true target even for relatively coarse discretizations. Appendix D.3 provides additional details.

These results clarify the practical role of bin construction. Equal-frequency bins stabilize empirical bin masses; empirical equal-width bins are sensitive to sample extremes when M has unbounded support; and predefined-width bins reflect externally fixed categories, such as clinically meaningful thresholds, but they lack the flexibility to adapt to

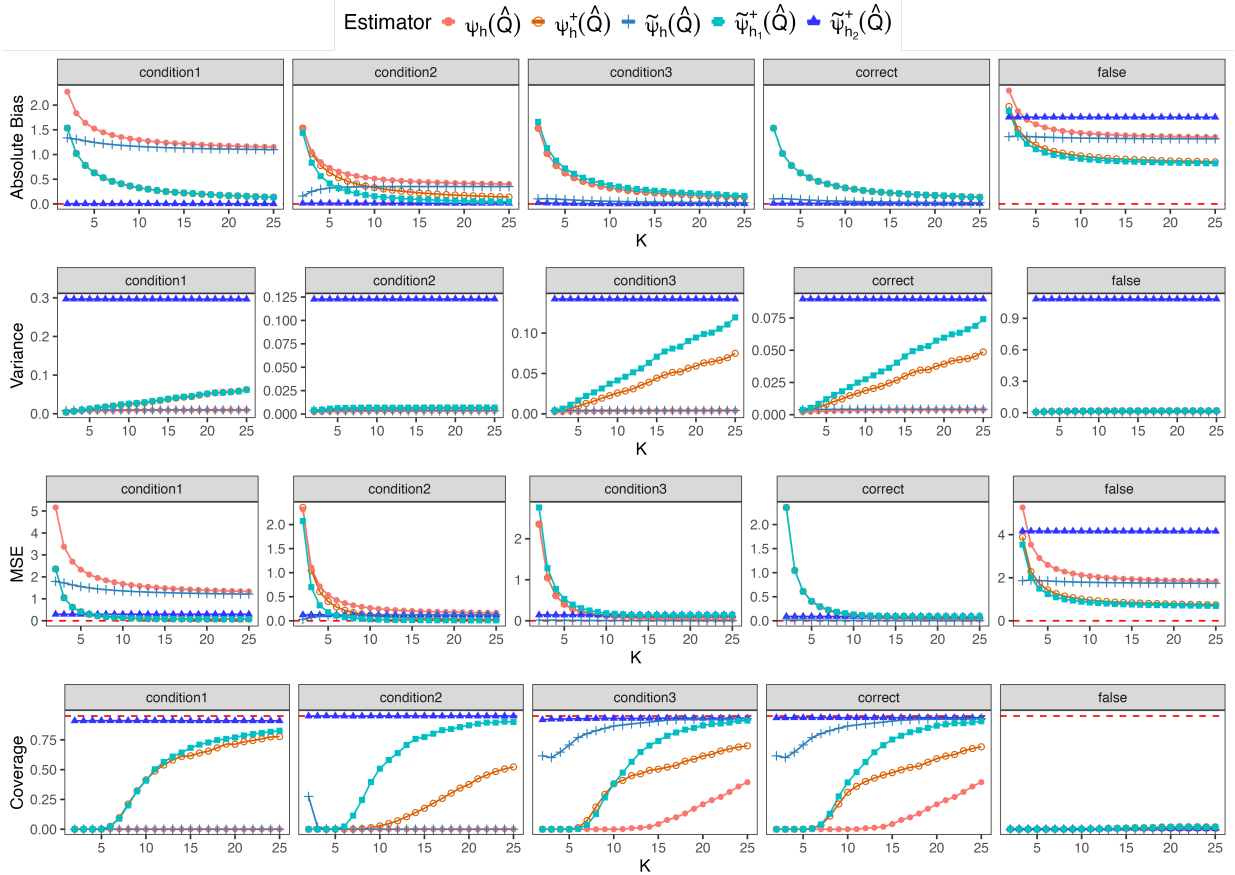


Figure 5: Comparison of plug-in and one-step estimators for $n = 5,000$ under nuisance-model misspecification. “Correct” and “False” indicate correctly specified and misspecified nuisance models, respectively. Conditions 1–3 correspond to misspecification of $\{\mu, \mu_k\}$, $\{g, g_k\}$, and π , respectively. Performance is summarized in terms of absolute bias, variance, MSE, and 95% confidence interval coverage.

a specific dataset, and coarsening error may remain even when K is large. The debiased estimator is considerably less sensitive to these choices, supporting its use when the discretization is dictated by interpretability, scientific convention, or data-analytic convenience.

We apply our method to the B_PROUD study, an observational evaluation of Mobile Stroke Unit dispatch and stroke outcomes among 768 reperfusion-eligible patients in Berlin, using the front-door functional. See Appendix E for details.

7 DISCUSSION

We study the bias induced by discretizing a continuous mediator in causal functionals. Even under correct identification and consistent nuisance estimation, naive discretization alters the target parameter and introduces a first-order coarsening error that scales with the bin width. Evaluating the outcome regression at within-bin means under the appropriate treatment-reference distribution removes this leading error term and reduces the approximation error to second or-

der under standard smoothness conditions. A key distinction is between functional approximation (coarsening) error and statistical estimation error. Influence-function-based estimators address the latter but do not correct the bias induced by discretization when the estimand itself has been modified. Correcting the functional is therefore essential; once this is done, standard semiparametric tools can be applied to control estimation error and conduct valid inference. In practice, discretization is often used to simplify computation or avoid direct conditional density estimation. We show that the resulting coarsening error can be formally characterized and substantially reduced through a simple within-bin correction that preserves computational tractability. Although we focus on mediation and front-door functionals, the same ideas extend to causal parameters involving integration of a regression surface against a conditional distribution under suitable smoothness conditions.

Acknowledgments

This work is supported by NIH grant R21ES036795.

References

- Chen Avin, Ilya Shpitser, and Judea Pearl. Identifiability of path-specific effects. In *Proceedings of the IJCAI*, pages 357–363, 12 2005.
- Rohit Bhattacharya, Razieh Nabi, and Ilya Shpitser. Semi-parametric inference for causal effects in graphical models with hidden variables. *Journal of Machine Learning Research*, 23:1–76, 2022.
- Mu-Chi Chung, Hui-Tsung Hsu, Yan-Chiao Mao, Chin-Ching Wu, Chih-Te Ho, Chiu-Shong Liu, and Chi-Jung Chung. Association and mediation analyses among multiple metals exposure, plasma folate, and community-based impaired estimated glomerular filtration rate in central taiwan. *Environmental Health*, 21(1):44, 2022.
- Katrina L Devick, Linda Valeri, Jarvis Chen, Alejandro Jara, Marie-Abèle Bind, and Brent A Coull. The role of body mass index at diagnosis of colorectal cancer on black–white disparities in survival: a density regression mediation approach. *Biostatistics*, 23(2):449–466, 2022.
- Martin Ebinger, Peter Harmel, Christian H Nolte, Ulrike Grittner, Bob Siegerink, and Heinrich J Audebert. Berlin prehospital or usual delivery of acute stroke care—study protocol. *International Journal of Stroke*, 12(6):653–658, 2017.
- Anna Guo and Razieh Nabi. Average causal effect estimation in dags with hidden variables: Beyond back-door and front-door criteria. *arXiv preprint arXiv:2409.03962*, 2024.
- Anna Guo, David Benkeser, and Razieh Nabi. Flexible nonparametric inference for causal effects under the front-door model. *arXiv preprint arXiv:2312.10234*, 2023.
- Anna Guo, Lin Liu, David Benkeser, and Razieh Nabi. Causal inference with the napkin graph. *arXiv preprint arXiv:2512.19861*, 2025.
- Ali Hemade and Pascale Salameh. Revisiting the obesity–anaemia paradox: Inflammation and iron homeostasis in the bmi–haemoglobin relationship. *Endocrinology, Diabetes & Metabolism*, 8(5):e70110, 2025.
- Guanglei Hong. Ratio of mediator probability weighting for estimating natural direct and indirect effects. *arXiv preprint arXiv:2506.03284*, 2025.
- Jiasheng Huang, Yehua Li, Angelique G Brellenthin, Duckchul Lee, Xuemei Sui, and Steven N Blair. Causal mediation analysis between resistance exercise and reduced risk of cardiovascular disease based on the aerobics center longitudinal study. *Journal of Applied Statistics*, 49(14): 3750–3767, 2022.
- Yi Li, Maya B Mathur, Daniel H Solomon, Paul M Ridker, Robert J Glynn, and Kazuki Yoshida. Effect measure modification by covariates in mediation: extending regression-based causal mediation analysis. *Epidemiology*, 34(5):661–672, 2023.
- Antonio R Linero. Simulation-based estimators of analytically intractable causal effects. *Biometrics*, 78(3): 1001–1017, 2022.
- Chang Liu and AmirEmad Ghassami. Two-stage nuisance function estimation for causal mediation analysis. *arXiv preprint arXiv:2404.00735*, 2024.
- Ruyi Liu, Liangyuan Hu, Francis Perry Wilson, Joshua L Warren, and Fan Li. A bayesian approach to the g-formula via iterative conditional regression. *Statistics in Medicine*, 44(13-14):e70123, 2025.
- Ze-wei Ma and Wei-nan Zeng. A multiple mediator model: Power analysis based on monte carlo simulation. *American Journal of Applied Psychology*, 3(3):72–79, 2014.
- Osvaldo F Morera, Mosi I Dane’el, Brandt A Smith, Alisha H Redelfs, Sarah L Ruiz, Kristopher J Preacher, and Leah D Whigham. Discretizing continuous variables in nutrition and obesity research: a practice that needs to be cut short. *Nutrition & Diabetes*, 13(1):20, 2023.
- Judea Pearl. Causal diagrams for empirical research. *Biometrika*, 82(4):669–688, 1995.
- Judea Pearl. Direct and indirect effects. In *Proceedings of the Seventeenth Conference on Uncertainty in Artificial Intelligence (UAI-01)*, pages 411–420. Morgan Kaufmann, San Francisco, 2001.
- Marco Piccininni, Tobias Kurth, Heinrich J Audebert, and Jessica L Rohmann. The effect of mobile stroke unit care on functional outcomes: an application of the front-door formula. *Epidemiology*, 34(5):712–720, 2023.
- Thomas S Richardson, Robin J Evans, James M Robins, and Ilya Shpitser. Nested markov properties for acyclic directed mixed graphs. *The Annals of Statistics*, 51(1): 334–361, 2023.
- Yasmmyn D Salinas, Zuoheng Wang, and Andrew T DeWan. Discovery and mediation analysis of cross-phenotype associations between asthma and body mass index in 12q13. 2. *American Journal of Epidemiology*, 190(1): 85–94, 2021.
- Ilya Shpitser. Counterfactual graphical models for longitudinal mediation analysis with unobserved confounding. *Cognitive science*, 37(6):1011–1035, 2013.
- Jin Tian and Judea Pearl. A general identification condition for causal effects. In *Eighteenth National Conference on Artificial Intelligence*, pages 567–573, 2002. ISBN 0-262-51129-0.

- Dustin Tingley, Teppei Yamamoto, Kentaro Hirose, Luke Keele, and Kosuke Imai. Mediation: R package for causal mediation analysis. *Journal of statistical software*, 59: 1–38, 2014.
- Mark J van der Laan and Daniel Rubin. Targeted maximum likelihood learning. *The International Journal of Biostatistics*, 2(1), 2006.
- Aad W Van der Vaart. *Asymptotic statistics*, volume 3. Cambridge university press, 2000.
- Tyler VanderWeele and Stijn Vansteelandt. Mediation analysis with multiple mediators. *Epidemiologic methods*, 2(1):95–115, 2014.
- Zeyi Wang, Lars van der Laan, Maya Petersen, Thomas Gerds, Kajsa Kvist, and Mark van der Laan. Targeted maximum likelihood based estimation for longitudinal mediation analysis. *Journal of Causal Inference*, 13(1): 20230013, 2025.
- Xiang Zhou. Semiparametric estimation for causal mediation analysis with multiple causally ordered mediators. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 84(3):794–821, 2022.

Coarsening Bias from Variable Discretization in Causal Functionals (Supplementary Material)

Xiaxian Ou¹

Razieh Nabi¹

¹Department of Biostatistics and Bioinformatics, Emory University, Atlanta, Georgia, USA

This Supplementary Material is organized as follows. Appendix A provides a glossary of the main terms and notation. Appendix B presents extensions to settings with multiple mediators. Appendix C contains all technical proofs. Appendix D provides additional details for the simulation studies. Appendix E reports results from the real data application.

A GLOSSARY OF TERMS AND NOTATION

Table S1 summarizes the main terms and notation used throughout the paper.

Table S1: Glossary of terms and notation.

Term or notation	Definition
Observed data and nuisance functions	
$O = (C, A, M, Y)$	Observed data: baseline covariates C , binary exposure A , mediator M , and outcome Y , respectively; a_0 and a_1 denote the two exposure levels.
$P_C, p(c)$	Marginal distribution and density of the baseline covariates C .
$\mu(m, a, c)$	Outcome regression, $\mathbb{E}(Y \mid M = m, A = a, C = c)$.
$\pi(a \mid c)$	Propensity score, $p(A = a \mid C = c)$.
$g(a \mid m, c)$	Generalized propensity score, $p(A = a \mid M = m, C = c)$.
Q	Collection of nuisance functions required by a target functional, such as μ , $f_{M A,C}$, and π .
Original target functionals	
$\theta(Q)(c)$	Conditional mediation functional, $\int \mu(m, a_1, c) f_{M A,C}(m \mid a_0, c) dm$.
$\psi(Q)$	Marginal mediation functional obtained by averaging $\theta(Q)(C)$ over P_C .
$\gamma(Q)$	Front-door functional.
Coarsening and discretized functionals	
$h, K, \mathcal{B}_k$	Coarsening map, number of bins, and the k th mediator bin, respectively.
$\widetilde{M} = h(M)$	Coarsened mediator taking values in $\{1, \dots, K\}$.
$w_k, w_{\max, K}$	Width of bin $\mathcal{B}_k$ and the maximum bin width.
$\mu_k(a, c)$	Coarsened-mediator outcome regression, $\mathbb{E}(Y \mid \widetilde{M} = k, A = a, C = c)$.
$\mu_{k,a_1}(a, c)$	Conditional mean of $\mu(M, a_1, c)$, $\mathbb{E}(\mu(M, a_1, c) \mid \widetilde{M} = k, A = a, C = c)$.
$m_k(a, c)$	Conditional mediator mean, $\mathbb{E}(M \mid \widetilde{M} = k, A = a, C = c)$.
$g_k(a, c)$	Conditional bin probability, $p(\widetilde{M} = k \mid A = a, C = c)$.
$p_{k,a}(m \mid c)$	Conditional mediator density, $p(m \mid \widetilde{M} = k, A = a, C = c)$.
$r_k(m \mid c)$	Within-bin conditional density ratio, $\frac{p_{k,a_0}(m \mid c)}{p_{k,a_1}(m \mid c)}$.
$\theta_h(Q)(c)$	Coarsened conditional mediation functional, $\sum_{k=1}^K \mu_k(a_1, c) g_k(a_0, c)$.
$\psi_h(Q), \gamma_h(Q)$	Coarsened versions of the marginal mediation and front-door functionals.

Continued on next page

Table S1 (continued)

Term or notation	Meaning
$\Delta_h(Q)(c)$	Coarsening error, $\theta_h(Q)(c) - \theta(Q)(c)$.
Debiasing and smoothing	
$\tilde{\theta}_h(Q)(c)$	Debiased coarsened functional, $\sum_{k=1}^K \mu\{m_k(a_0, c), a_1, c\} g_k(a_0, c)$.
$\tilde{\psi}_h(Q), \tilde{\gamma}_h(Q)$	Debiased coarsened versions of the marginal mediation and front-door functionals.
$R_k(c)$	Within-bin debiasing remainder, $\mu\{m_k(a_0, c), a_1, c\} - \mu_{k, a_1}(a_0, c)$.
$\tilde{\Delta}_h(Q)(c)$	Approximation error of the debiased coarsened functional, $\tilde{\theta}_h(Q)(c) - \theta(Q)(c)$.
$\mathcal{K}, b, \mathcal{K}_b$	Kernel function, bandwidth, and rescaled kernel $\mathcal{K}_b(u) = b^{-1} \mathcal{K}(u/b)$.
$\omega_{b,k}(m a_1, c)$	Normalized localized kernel weight centered at $m_k(a_0, c)$.
$\mu_{b,k}(a_1, c)$	Kernel-localized outcome mean for the k th bin.
$\tilde{\theta}_{h,b}(Q)(c)$	Smoothed, debiased coarsened functional, $\sum_{k=1}^K \mu_{b,k}(a_1, c) g_k(a_0, c)$.
$\tilde{\psi}_{h,b}(Q), \tilde{\gamma}_{h,b}(Q)$	Smoothed, debiased versions of the marginal mediation and front-door functionals.
$\tilde{\Delta}_{h,b}^s(Q)(c)$	Smoothing error, $\tilde{\theta}_{h,b}(Q)(c) - \tilde{\theta}_h(Q)(c)$.
$\tilde{\Delta}_{h,b}(Q)(c)$	Total approximation error, $\tilde{\theta}_{h,b}(Q)(c) - \theta(Q)(c)$.
Estimators	
$\theta(\hat{Q})(c)$	Plug-in estimator of the original conditional mediation functional.
$\psi(\hat{Q}), \gamma(\hat{Q})$	Plug-in estimators of the original mediation and front-door functionals.
$\theta_h(\hat{Q})(c)$	Coarsened plug-in estimator, $\sum_{k=1}^K \hat{\mu}_k(a_1, c) \hat{g}_k(a_0, c)$.
$\psi_h(\hat{Q}), \gamma_h(\hat{Q})$	Coarsened plug-in estimators obtained by replacing $\theta(\hat{Q})$ with $\theta_h(\hat{Q})$.
$\tilde{\theta}_h(\hat{Q})(c)$	Debiased coarsened plug-in estimator, $\sum_{k=1}^K \hat{\mu}\{\hat{m}_k(a_0, c), a_1, c\} \hat{g}_k(a_0, c)$.
$\tilde{\psi}_h(\hat{Q}), \tilde{\gamma}_h(\hat{Q})$	Debiased coarsened plug-in estimators obtained by replacing $\theta(\hat{Q})$ with $\tilde{\theta}_h(\hat{Q})$.
$\tilde{\theta}_{h,b}(\hat{Q})(c)$	Smoothed, debiased coarsened plug-in estimator based on $\hat{\mu}_{b,k}$ and $\hat{g}_k$.
$\tilde{\psi}_{h,b}^{\text{fixed}}(\hat{Q}), \tilde{\psi}_{h,b}^{\text{one-step}}(\hat{Q})$	Fixed-center plug-in and one-step estimators of the smoothed, debiased mediation functional.
$\psi^+(\hat{Q}), \psi_h^+(\hat{Q})$	One-step estimators targeting the original and coarsened mediation functionals, respectively.
$\tilde{\psi}_{h_1}^+(\hat{Q})$	Derivative one-step estimator that replaces $\hat{\mu}_k$ by $\hat{\mu}(m_k(a_0, c), a_1, c)$ and $\theta_h(\hat{Q})(c)$ with $\tilde{\theta}_h(\hat{Q})(c)$ in $\psi_h^+(\hat{Q})$.
$\tilde{\psi}_{h_2}^+(\hat{Q})$	Derivative one-step estimator that replaces $\theta(\hat{Q})(c)$ with $\tilde{\theta}_h(\hat{Q})(c)$ in $\psi^+(\hat{Q})$.
$\gamma_s(\hat{Q})$	Sequential-regression plug-in estimator of the front-door functional.

B EXTENSIONS TO MULTIPLE MEDIATORS

We discuss the extensions with two (sets of) mediators. Let M_1 and M_2 denote two ordered continuous or multi-dimensional mediators. Define the outcome regression $\mu(m_1, m_2, a, c) = \mathbb{E}(Y | M_1 = m_1, M_2 = m_2, A = a, C = c)$, the conditional mediator density $f_{M_1|A,C}(m_1 | a, c) = p(M_1 = m_1 | A = a, C = c)$ and $f_{M_2|M_1,A,C}(m_2 | m_1, a, c) = p(M_2 = m_2 | M_1 = m_1, A = a, C = c)$. Let $Q = \{\mu, f_{M_1|A,C}, f_{M_2|M_1,A,C}\}$ collects the nuisance functions. Let $(a_y, a_1, a_2) \in \{0, 1\}^3$, and for each $c \in \mathcal{C}$,

$$\theta(Q)(c) = \iint \mu(m_2, m_1, a_y, c) f_{M_2|M_1,A,C}(m_2 | m_1, a_2, c) f_{M_1|A,C}(m_1 | a_1, c) dm_2 dm_1. \quad (\text{S1})$$

Similarly, for $i = 1, 2$, let $h_i : \mathbb{R} \mapsto \{1, \dots, K_i\}$ denote the measurable discretization maps that partitions the support of M_i into K_i disjoint bins: $\mathcal{B}_{i,k_i} = \{m_i : h_i(m_i) = k_i\}$, $k_i \in \{1, \dots, K_i\}$. Let $w_{i,k_i} = \sup_{m_{i,1}, m_{i,2} \in \mathcal{B}_{i,k_i}} |m_{i,1} - m_{i,2}|$ denote the bin width and define the coarsened mediator $\tilde{M}_i = h_i(M_i)$, $i = 1, 2$.

For each bin $k_i \in \{1, \dots, K_i\}$, define $g_{k_1}(a_1, c) = p(\tilde{M}_1 = k_1 | A = a_1, C = c)$, $g_{k_1, k_2}(a_2, c) = p(\tilde{M}_2 = k_2 | \tilde{M}_1 = k_1, A = a_2, C = c)$, and $\mu_{k_1, k_2}(a_y, c) = \mathbb{E}(Y | \tilde{M}_2 = k_2, \tilde{M}_1 = k_1, A = a_y, C = c)$. The coarsened analogue of (S1) is:

$$\theta_h(Q)(c) = \sum_{k_1=1}^{K_1} \sum_{k_2=1}^{K_2} \mu_{k_1, k_2}(a_y, c) g_{k_1, k_2}(a_2, c) g_{k_1}(a_1, c). \quad (\text{S2})$$

To introduce the similar correction as one mediator in main manuscript, we first define

$$\begin{aligned}
g_{k_2}(m_1, a_2, c) &= p(\widetilde{M}_2 = k_2 | M_1 = m_1, A = a_2, C = c), \\
\mu_{k_2, a_y}(m_1, a_2, c) &= \mathbb{E}(\mu(M_1, M_2, a_y, C) | \widetilde{M}_2 = k_2, M_1 = m_1, A = a_2, C = c), \\
\eta_{k_2, a_y}(m_1, a_2, c) &= \mu_{k_2, a_y}(m_1, a_2, c) p(\widetilde{M}_2 = k_2 | M_1 = m_1, A = a_2, C = c), \\
\mu_{\eta, k_1, k_2, a_2, a_y}(a_1, c) &= \mathbb{E}(\eta_{k_2, a_y}(M_1, a_2, C) | \widetilde{M}_1 = k_1, A = a_1, C = c).
\end{aligned}$$

Then, the conditional functional $\theta(Q)(c)$ could be written as

$$\begin{aligned}
\theta(Q)(c) &= \sum_{k_1=1}^{K_1} \sum_{k_2=1}^{K_2} \left\{ \int \int \mu(m_1, m_2, a_y, c) p(m_2 | \widetilde{M}_2 = k_2, m_1, a_2, c) dm_2 \right\} \\
&\quad \times p(\widetilde{M}_2 = k_2 | m_1, a_2, c) p(m_1 | \widetilde{M}_1 = k_1, a_1, c) dm_1 \Big\} \\
&\quad \times p(\widetilde{M}_1 = k_1 | a_1, c) \\
&= \sum_{k_1=1}^{K_1} \sum_{k_2=1}^{K_2} \mu_{\eta, k_1, k_2, a_2, a_y}(a_1, c) g_{k_1}(a_1, c). \tag{S3}
\end{aligned}$$

Define the within-bin mean

$$\begin{aligned}
m_{1, k_1}(a_1, c) &= \mathbb{E}[M_1 | \widetilde{M}_1 = k_1, A = a_1, C = c], \\
m_{2, k_2}(m_1, a_2, c) &= \mathbb{E}[M_2 | \widetilde{M}_2 = k_2, M_1 = m_1, A = a_2, C = c].
\end{aligned}$$

Thus, we define the debiased coarsened functional as

$$\widetilde{\theta}_h(Q)(c) = \sum_{k_1=1}^{K_1} \sum_{k_2=1}^{K_2} \mu(m_{1, k_1}(a_1, c), m_{2, k_2}(m_{1, k_1}(a_1, c), a_2, c), a_y, c) g_{k_2}(m_{1, k_1}(a_1, c), a_2, c) g_{k_1}(a_1, c). \tag{S4}$$

For each $c \in \mathcal{C}$, the coarsening error $\widetilde{\Delta}_h(Q)(c) = \widetilde{\theta}_h(Q)(c) - \theta(Q)(c)$ satisfies

$$\begin{aligned}
\widetilde{\Delta}_h(Q)(c) &= \sum_{k_1=1}^{K_1} \sum_{k_2=1}^{K_2} \left\{ \mu(m_{1, k_1}(a_1, c), m_{2, k_2}(m_{1, k_1}(a_1, c), a_2, c), a_y, c) g_{k_2}(m_{1, k_1}(a_1, c), a_2, c) \right. \\
&\quad \left. - \mu_{\eta, k_1, k_2, a_2, a_y}(a_1, c) \right\} g_{k_1}(a_1, c). \tag{S5}
\end{aligned}$$

Assume $\mu(m_1, m_2, a_y, c)$ is twice continuously differentiable on each $\mathcal{B}_{i, k_i}$. Inside each bin $\mathcal{B}_{2, k_2}$, apply a second-order Taylor expansion of $\mu(m_1, m_2, a_y, c)$ around $m_{2, k_2}(m_1, a_2, c)$:

$$\begin{aligned}
\mu(m_1, M_2, a_y, c) &= \mu(m_1, m_{2, k_2}(m_1, a_2, c), a_y, c) + \mu'_{m_2}(m_1, m_{2, k_2}(m_1, a_2, c), a_y, c)(M_2 - m_{2, k_2}(m_1, a_2, c)) \\
&\quad + \frac{1}{2} \mu''_{m_2}(m_1, M_2^*, a_y, c)(M_2 - m_{2, k_2}(m_1, a_2, c))^2, \tag{S6}
\end{aligned}$$

for some $M_2^* \in \mathcal{B}_{2, k_2}$.

Assume $\eta_{k_2, a_y}(m_1, a_2, c)$ is twice continuously differentiable on each $\mathcal{B}_{1, k_1}$. Inside each bin $\mathcal{B}_{1, k_1}$ for M_1 , apply a second-order Taylor expansion of $\eta_{k_2, a_y}(M_1, a_2, c)$ around $m_{1, k_1}(a_1, c)$:

$$\begin{aligned}
\eta_{k_2, a_y}(M_1, a_2, c) &= \eta_{k_2, a_y}(m_{1, k_1}(a_1, c), a_2, c) + (\eta_{k_2, a_y})'_{m_1}(m_{1, k_1}(a_1, c), a_2, c)(M_1 - m_{1, k_1}(a_1, c)) \\
&\quad + \frac{1}{2} (\eta_{k_2, a_y})''_{m_1}(M_1^*, a_2, c)(M_1 - m_{1, k_1}(a_1, c))^2. \tag{S7}
\end{aligned}$$

for some $M_1^* \in \mathcal{B}_{1, k_1}$.

Take conditional expectation sequentially :

$$\begin{aligned}
& \mu_{\eta,k_1,k_2,a_2,a_y}(a_1, c) \\
&= \mathbb{E}(\eta_{k_2,a_y}(M_1, a_2, c) \mid \widetilde{M}_1=k_1, A=a_1, C=c) \\
&\stackrel{S7}{=} \eta_{k_2,a_y}(m_{1,k_1}(a_1, c), a_2, c) + \frac{1}{2} \mathbb{E}((\eta_{k_2,a_y})''_{m_1}(M_1^*, a_2, c)(M_1 - m_{1,k_1}(a_1, c))^2 \mid \widetilde{M}_1=k_1, A=a_1, C=c) \\
&= \mu_{k_2,a_y}(m_{1,k_1}(a_1, c), a_2, c) g_{k_2}(m_{1,k_1}(a_1, c), a_2, c) \\
&\quad + \frac{1}{2} \mathbb{E}((\eta_{k_2,a_y})''_{m_1}(M_1^*, a_2, c)(M_1 - m_{1,k_1}(a_1, c))^2 \mid \widetilde{M}_1=k_1, A=a_1, C=c) \\
&= \mathbb{E}(\mu(m_{1,k_1}(a_1, c), M_2, a_y, c) \mid \widetilde{M}_2=k_2, m_{1,k_1}(a_1, c), A=a_2, C=c) g_{k_2}(m_{1,k_1}(a_1, c), a_2, c) \\
&\quad + \frac{1}{2} \mathbb{E}((\eta_{k_2,a_y})''_{m_1}(M_1^*, a_2, c)(M_1 - m_{1,k_1}(a_1, c))^2 \mid \widetilde{M}_1=k_1, A=a_1, C=c) \\
&\stackrel{S6}{=} \left\{ \mu(m_{1,k_1}(a_1, c), m_{2,k_2}(m_{1,k_1}(a_1, c), a_2, c), a_y, c) \right. \\
&\quad + \frac{1}{2} \mathbb{E}(\mu''_{m_2}(m_{1,k_1}(a_1, c), M_2^*, a_y, c)(M_2 - m_{2,k_2}(m_{1,k_1}(a_1, c), a_2, c))^2 \mid \widetilde{M}_2=k_2, m_{1,k_1}(a_1, c), A=a_2, C=c) \Big\} \\
&\quad \times g_{k_2}(m_{1,k_1}(a_1, c), a_2, c) \\
&\quad + \frac{1}{2} \mathbb{E}((\eta_{k_2,a_y})''_{m_1}(M_1^*, a_2, c)(M_1 - m_{1,k_1}(a_1, c))^2 \mid \widetilde{M}_1=k_1, A=a_1, C=c) .
\end{aligned}$$

Thus

$$\begin{aligned}
R_{k_1,k_2}(c) &= \mu(m_{1,k_1}(a_1, c), m_{2,k_2}(m_{1,k_1}(a_1, c), a_2, c), a_y, c) g_{k_2}(m_{1,k_1}(a_1, c), a_2, c) - \mu_{\eta,k_1,k_2,a_2,a_y}(a_1, c) \\
&= -\frac{1}{2} \mathbb{E}(\mu''_{m_2}(m_{1,k_1}(a_1, c), M_2^*, a_y, c)(M_2 - m_{2,k_2}(m_{1,k_1}(a_1, c), a_2, c))^2 \mid \widetilde{M}_2=k_2, m_{1,k_1}(a_1, c), A=a_2, C=c) \\
&\quad \times g_{k_2}(m_{1,k_1}(a_1, c), a_2, c) \\
&\quad - \frac{1}{2} \mathbb{E}((\eta_{k_2,a_y})''_{m_1}(M_1^*, a_2, c)(M_1 - m_{1,k_1}(a_1, c))^2 \mid \widetilde{M}_1=k_1, A=a_1, C=c) .
\end{aligned}$$

Given $|g_{k_2}(m_{1,k_1}(a_1, c), a_2, c)| < 1$, the monotonicity and linearity of expectations, taking absolute values, we obtain the following upper bound on $|R_{k_1,k_2}(c)|$:

$$\begin{aligned}
|R_{k_1,k_2}(c)| &\leq \frac{1}{2} \mathbb{E}(|\mu''_{m_2}(m_{1,k_1}(a_1, c), M_2^*, a_y, c)|(M_2 - m_{2,k_2}(m_{1,k_1}(a_1, c), a_2, c))^2 \mid \widetilde{M}_2=k_2, m_{1,k_1}(a_1, c), A=a_2, C=c) \\
&\quad + \frac{1}{2} \mathbb{E}(|(\eta_{k_2,a_y})''_{m_1}(M_1^*, a_2, c)|(M_1 - m_{1,k_1}(a_1, c))^2 \mid \widetilde{M}_1=k_1, A=a_1, C=c) \\
&\leq \frac{1}{2} \sup_{m_2 \in \mathcal{B}_{2,k_2}(c)} |\mu''_{m_2}(m_{1,k_1}(a_1, c), m_2, a_y, c)| \mathbb{E}((M_2 - m_{2,k_2}(m_{1,k_1}(a_1, c), a_2, c))^2 \mid \widetilde{M}_2=k_2, m_{1,k_1}(a_1, c), A=a_2, C=c) \\
&\quad + \frac{1}{2} \sup_{m_1 \in \mathcal{B}_{1,k_1}(c)} |(\eta_{k_2,a_y})''_{m_1}(m_1, a_2, c)| \mathbb{E}((M_1 - m_{1,k_1}(a_1, c))^2 \mid \widetilde{M}_1=k_1, A=a_1, C=c) .
\end{aligned}$$

For any (a_1, a_2, c) , the conditional law $p(m_1 \mid \widetilde{M}_1 = k_1, A = a_1, C = c)$ is supported on $B_{1,k_1}(c)$ and $p(m_2 \mid \widetilde{M}_2 = k_2, M_1 = m_{1,k_1}(a_1, c), A = a_2, C = c)$ is supported on $B_{2,k_2}(c)$, hence

$$\begin{aligned}
\text{Var}(M_1 \mid k_1, a_1, c) &\leq \frac{(\sup_{m_{1,1}, m_{1,2} \in \mathcal{B}_{1,k_1}(c)} |m_{1,1} - m_{1,2}|)^2}{4} = \frac{w_{1,k_1}(c)^2}{4} \\
\text{Var}(M_2 \mid k_2, m_{1,k_1}(a_1, c), a_2, c) &\leq \frac{(\sup_{m_{2,1}, m_{2,2} \in \mathcal{B}_{2,k_2}(c)} |m_{2,1} - m_{2,2}|)^2}{4} = \frac{w_{2,k_2}(c)^2}{4} .
\end{aligned}$$

For $\sup_{m_1 \in \mathcal{B}_{1,k_1}(c)} |(\eta_{k_2,a_y})''_{m_1}(m_1, a_2, c)| \leq L_1(c)$ and $\sup_{m_2 \in \mathcal{B}_{2,k_2}(c)} |\mu''_{m_2}(m_{1,k_1}(a_1, c), m_2, a_y, c)| \leq L_2(c)$, and the fact that $\sum_{k=1}^K g_{k_1}(a_1, c) = 1$, we have $\widetilde{\Delta}_h(Q)(c) = O(w_{\max,1,K_1}(c)^2 + w_{\max,2,K_2}(c)^2)$.

C PROOFS

C.1 IDENTIFICATION PROOFS

For the identification of mediation functional $\mathbb{E}(Y(a_1, M(a_0)))$ and the front-door functional $\mathbb{E}(Y(a_0))$, we first introduce general assumptions that apply throughout. For any a and m , we assume

- (A) *Consistency*, which indicates that observed outcome and mediators match their counterfactuals when treatment and mediator values are set at observed values; i.e. $Y(a, m) = Y$ if $A = a$ and $M = m$, and $M(a) = M$ if $A = a$.
- (B) *Positivity*, which states that $P(A = 1 | C = c) > 0$ when $P(C = c) > 0$, and $P(A = 1 | M = m, C = c) > 0$ when $P(M = m, C = c) > 0$.

To identify the mediation functional $\mathbb{E}(Y(a_1, M(a_0)))$, we additionally need:

- (C1) *Conditional ignorability*, which assumes the absence of unmeasured confounders between the treatment-mediator, treatment-outcome and mediator-outcome pairs; i.e. for any a, a_0, a_1 and m : (i) $Y(a_1, m), M(a_0) \perp\!\!\!\perp A | C$; (ii) $Y(a_1, m) \perp\!\!\!\perp M(a) | A, C$.

Under (A)–(C1), the mediation functional is identified as

$$\begin{aligned}
 & \mathbb{E}(Y(a_1, M(a_0))) \\
 &= \iint \mathbb{E}(Y(a_1, m) | M(a_0) = m, C = c) p(M(a_0) = m | C = c) p(C = c) dm dc \\
 &= \iint \mathbb{E}(Y(a_1, m) | M(a_0) = m, A = a_1, C = c) p(M = m | A = a_0, C = c) p(C = c) dm dc \\
 &= \iint \mathbb{E}(Y(a_1, m) | M(a_1) = m, A = a_1, C = c) p(M = m | A = a_0, C = c) p(C = c) dm dc \\
 &= \iint \mathbb{E}(Y | M = m, A = a_1, C = c) p(M = m | A = a_0, C = c) p(C = c) dm dc
 \end{aligned} \tag{S8}$$

To identify the front-door functional $\mathbb{E}(Y(a_0))$, in addition to (A) and (B), we require:

- (C2) *Conditional ignorability*, which assumes the absence of unmeasured confounders between the treatment-mediator, and mediator-outcome pairs; i.e. (i) $M(a_0) \perp\!\!\!\perp A | C$; (ii) $Y(m) \perp\!\!\!\perp M | A, C$; (iii) $Y(m) \perp\!\!\!\perp M(a_0) | C$.
- (D) *No direct effect*, which assumes that M blocks all directed paths from A to Y , i.e., $Y(a, m) = Y(m)$ for any a and m .

Under (A), (B), (C2) and (D), the front-door functional is identified as

$$\begin{aligned}
 & \mathbb{E}(Y(a_0)) \\
 &= \iint \mathbb{E}(Y(a_0) | M(a_0) = m, C = c) p(M(a_0) = m | C = c) p(C = c) dm dc \\
 &= \iint \mathbb{E}(Y(a_0, m) | M(a_0) = m, C = c) p(M(a_0) = m | A = a_0, C = c) p(C = c) dm dc \\
 &= \iint \mathbb{E}(Y(m) | M(a_0) = m, C = c) p(M = m | A = a_0, C = c) p(C = c) dm dc \\
 &= \iint \mathbb{E}(Y(m) | C = c) p(M = m | A = a_0, C = c) p(C = c) dm dc \\
 &= \iint \left\{ \sum_{a \in \{0,1\}} \mathbb{E}(Y | M = m, A = a, C = c) p(A = a | C = c) \right\} p(M = m | A = a_0, C = c) p(C = c) dm dc \\
 &= p(A = a_0) \mathbb{E}(Y | A = a_0) + \iint \mathbb{E}(Y | M = m, A = a_1, C = c) p(M = m | A = a_0, C = c) p(A = a_1, C = c) dm dc.
 \end{aligned} \tag{S9}$$

C.2 LEMMA 3.1

Given our notations, we can rewrite $\theta(Q)(C)$ in (1) as:

$$\theta(Q)(C) = \sum_{k=1}^K \left\{ \int \mu(m, a_1, c) p(m \mid \widetilde{M} = k, a_0, c) dm \right\} g_k(a_0, c) = \sum_{k=1}^K \mu_{k,a_1}(a_0, c) g_k(a_0, c), \quad (\text{S10})$$

By definition of the coarsened estimand,

$$\theta_h(Q)(c) = \sum_{k=1}^K \mu_k(a_1, c) g_k(a_0, c).$$

Therefore,

$$\Delta_h(Q)(c) = \theta_h(Q)(c) - \theta(Q)(c) \quad (\text{S11})$$

$$= \sum_{k=1}^K \mu_k(a_1, c) g_k(a_0, c) - \sum_{k=1}^K \mu_{k,a_1}(a_0, c) g_k(a_0, c) \quad (\text{S12})$$

$$= \sum_{k=1}^K \{ \mu_k(a_1, c) - \mu_{k,a_1}(a_0, c) \} g_k(a_0, c), \quad (\text{S13})$$

which proves (6).

Further, suppose that $\sup_{m \in \mathcal{B}_k(c)} |\mu'_m(m, a_1, c)| \leq L(c)$ for some square-integrable function under P_C . Then from (6),

$$|\Delta_h(Q)(c)| \leq \sum_{k=1}^K |\mu_k(a_1, c) - \mu_{k,a_1}(a_0, c)| g_k(a_0, c) \quad (\text{S14})$$

$$\leq L(c) \sum_{k=1}^K |m_k(a_1, c) - m_k(a_0, c)| g_k(a_0, c). \quad (\text{S15})$$

Since both conditional means $m_k(a_1, c)$ and $m_k(a_0, c)$ lie in the same bin $B_k(c)$, their difference is bounded by the bin width,

$$|m_k(a_1, c) - m_k(a_0, c)| \leq w_k(c),$$

so

$$|\Delta_h(Q)(c)| \leq L(c) \sum_{k=1}^K w_k(c) g_k(a_0, c). \quad (\text{S16})$$

The weights $g_k(a_0, c)$ form a probability distribution over k , hence $\sum_{k=1}^K w_k(c) g_k(a_0, c)$ is a weighted average of the bin widths and is bounded by the maximum bin width,

$$\sum_{k=1}^K w_k(c) g_k(a_0, c) \leq \max_k w_k(c) =: w_{\max, K}(c).$$

Therefore

$$\Delta_h(Q)(c) = O(w_{\max, K}(c)).$$

When the mediator has bounded support and the K bins are chosen to have equal width, $w_{\max, K}(c) = O(1/K)$, so $\Delta_h(Q)(c) = O(1/K)$.

To prove (8), we assume μ is twice continuously differentiable and define the within-bin mean

$$m_k(a, c) = \mathbb{E}(M \mid A = a, C = c, \widetilde{M} = k). \quad (\text{S17})$$

Applying a first-order Taylor expansion of $\mu(M, a_1, c)$ around $m_k(a, c)$ yields

$$\mu(M, a_1, c) \approx \mu(m_k(a, c), a_1, c) + \mu'_m(m_k(a, c), a_1, c) (M - m_k(a, c)) ,$$

where μ'_m denotes the partial derivative of μ with respect to m . Taking expectations w.r.t. $p(m | \widetilde{M} = k, a, c)$ shows that $\mathbb{E}(\mu(M, a_1, c) | \widetilde{M} = k, a, c) \approx \mu(m_k(a, c), a_1, c)$ up to second-order remainder terms. Hence, $\mu_k(a_1, c) - \mu_{k,a_1}(a_0, c)$ can be approximated by

$$\begin{aligned} \mu_k(a_1, c) - \mu_{k,a_1}(a_0, c) &\approx \mu(m_k(a_1, c), a_1, c) - \mu(m_k(a_0, c), a_1, c) \\ &\approx \mu'_m(m_k(a_1, c), a_1, c) (m_k(a_1, c) - m_k(a_0, c)) , \end{aligned} \quad (\text{S18})$$

where the second line follows another first-order Taylor expansion of μ around $m_k(a_1, c)$.

C.3 REMARK 3.2

For each bin index k and covariate value c , define

$$p_{k,a}(m | c) = p(m | \widetilde{M} = k, A = a, C = c) , \quad r_k(m | c) = \frac{p_{k,a_0}(m | c)}{p_{k,a_1}(m | c)} . \quad (\text{S19})$$

We can write

$$\begin{aligned} \mu_{k,a_1}(a_0, c) - \mu_k(a_1, c) &= \int \mu(m, a_1, c) p_{k,a_0}(m | c) dm - \int \mu(m, a_1, c) p_{k,a_1}(m | c) dm \\ &= \int \mu(m, a_1, c) \{r_k(m | c) - 1\} p_{k,a_1}(m | c) dm \\ &= \mathbb{E}_{p_{k,a_1}} \left(\mu(M, a_1, c) (r_k(M | c) - 1) \right) \\ &= \mathbb{E}_{p_{k,a_1}} \left((\mu(M, a_1, c) - \mathbb{E}_{p_{k,a_1}}(\mu(M, a_1, c))) (r_k(M | c) - 1) \right) \\ &= \text{Cov}_{p_{k,a_1}}(\mu(M, a_1, c), r_k(M | c)) , \end{aligned}$$

since $\mathbb{E}_{p_{k,a_1}}(r_k(M | c)) = 1$.

C.4 LEMMA 4.1

In the proof of Lemma 3.1, we showed $\theta(Q)(c) = \sum_{k=1}^K \mu_{k,a_1}(a_0, c) g_k(a_0, c)$. By definition of the debiased coarsened functional, $\widetilde{\theta}_h(Q)(c) = \sum_{k=1}^K \mu(m_k(a_0, c), a_1, c) g_k(a_0, c)$. Subtracting the two displays yields (12).

We now bound $R_k(c)$ for a fixed bin index k . Inside each bin $\mathcal{B}_k$, apply a second-order Taylor expansion of $\mu(M, a_1, c)$ around $m_k(a_0, c)$:

$$\mu(M, a_1, c) = \mu(m_k(a_0, c), a_1, c) + \mu'_m(m_k(a_0, c), a_1, c)(M - m_k(a_0, c)) + \frac{1}{2} \mu''_m(M^*, a_1, c)(M - m_k(a_0, c))^2 ,$$

for some $M^* \in \mathcal{B}_k$ (by the mean-value form of the remainder). Taking the conditional expectation under $(\widetilde{M}=k, A=a_0, C=c)$ gives

$$\mathbb{E}(\mu(M, a_1, c) | k, a_0, c) = \mu(m_k(a_0, c), a_1, c) + \frac{1}{2} \mathbb{E}(\mu''_m(M^*, a_1, c)(M - m_k(a_0, c))^2 | k, a_0, c) .$$

Thus

$$R_k(c) = \mu(m_k(a_0, c), a_1, c) - \mu_{k,a_1}(a_0, c) = -\frac{1}{2} \mathbb{E}(\mu''_m(M^*, a_1, c)(M - m_k(a_0, c))^2 | k, a_0, c) .$$

Taking absolute values and given the monotonicity and linearity of expectations, we obtain the following upper bound on $|R_k(c)|$:

$$\begin{aligned} |R_k(c)| &\leq \frac{1}{2} \mathbb{E}(|\mu''_m(M^*, a_1, c)| (M - m_k(a_0, c))^2 | k, a_0, c) \\ &\leq \frac{1}{2} \sup_{m \in \mathcal{B}_k(c)} |\mu''_m(m, a_1, c)| \mathbb{E}((M - m_k(a_0, c))^2 | k, a_0, c) . \end{aligned}$$

Finally, since the conditional law of M given $(A = a_0, C = c, \widetilde{M} = k)$ is supported on $\mathcal{B}_k$, its range is at most w_k , and hence

$$\begin{aligned}\mathbb{E}\left((M - m_k(a_0, c))^2 \mid A = a_0, C = c, \widetilde{M} = k\right) &= \text{Var}(M \mid A = a_0, C = c, \widetilde{M} = k) \\ &\leq \frac{(\sup_{m_1, m_2 \in \mathcal{B}_k} |m_1 - m_2|)^2}{4} \\ &= w_k^2/4\end{aligned}$$

Combining with $\sup_{m \in \mathcal{B}_k} |\mu''_m(m, a_1, c)| \leq L(c)$ yields $|R_k(c)| \leq L(c) w_k^2/8$. Furthermore, using the fact that $\sum_{k=1}^K g_k(a_0, c) = 1$, we obtain

$$|\widetilde{\Delta}_h(Q)(c)| \leq \sum_{k=1}^K |R_k(c)| g_k(a_0, c) \leq \frac{L(c)}{8} \sum_{k=1}^K w_k^2 g_k(a_0, c) \leq \frac{L(c)}{8} w_{\max, K}^2,$$

which implies $\widetilde{\Delta}_h(Q)(c) = O(w_{\max, K}^2)$.

C.5 LEMMA 5.1

By definition:

$$\widetilde{\theta}_{h,b}(Q)(c) = \sum_{k=1}^K \mu_{b,k}(a_1, c) g_k(a_0, c), \quad \widetilde{\theta}_h(Q)(c) = \sum_{k=1}^K \mu(m_k(a_0, c), a_1, c) g_k(a_0, c),$$

so

$$\begin{aligned}\widetilde{\Delta}_{h,b}^s(Q)(c) &= \widetilde{\theta}_{h,b}(Q)(c) - \widetilde{\theta}_h(Q)(c) \\ &= \sum_{k=1}^K \{\mu_{b,k}(a_1, c) - \mu(m_k(a_0, c), a_1, c)\} g_k(a_0, c),\end{aligned}$$

which establishes the first display.

It remains to show that each term in braces is of order b^2 uniformly in k, c under the stated conditions. Fix k and c ,

$$\begin{aligned}\mu_{b,k}(a_1, c) &= E\{\mu(m, a_1, c) \omega_{b,k}(M \mid a_1, c) \mid A = a_1, C = c\} \\ &= \frac{\mathbb{E}\{\mu(m, a_1, c) \mathcal{K}_b(M - m_k(a_0, c)) \mid A = a_1, C = c\}}{\mathbb{E}\{\mathcal{K}_b(M - m_k(a_0, c)) \mid A = a_1, C = c\}} \\ &= \frac{N_b}{D_b},\end{aligned}$$

where

$$\begin{aligned}N_b &= \int \mu(m, a_1, c) \mathcal{K}_b(m - m_k(a_0, c)) f_{M \mid A, C}(m \mid a_1, c) dm, \\ D_b &= \int \mathcal{K}_b(m - m_k(a_0, c)) f_{M \mid A, C}(m \mid a_1, c) dm.\end{aligned}$$

Make the change of variables $m = m_k(a_0, c) + bu$, so that $dm = b du$ and $\mathcal{K}_b(m - m_k(a_0, c)) = \frac{1}{b} \mathcal{K}\left(\frac{m - m_k(a_0, c)}{b}\right) = \frac{1}{b} \mathcal{K}(u)$. Then

$$\begin{aligned}N_b &= \int \mu(m_k(a_0, c) + bu, a_1, c) \mathcal{K}_b(m_k(a_0, c) + bu - m_k(a_0, c)) f_{M \mid A, C}(m_k(a_0, c) + bu \mid a_1, c) b du \\ &= \int \mu(m_k(a_0, c) + bu, a_1, c) f_{M \mid A, C}(m_k(a_0, c) + bu \mid a_1, c) \mathcal{K}(u) du, \\ D_b &= \int f_{M \mid A, C}(m_k(a_0, c) + bu \mid a_1, c) \mathcal{K}(u) du.\end{aligned}$$

By assumption (i), $\mu(\cdot)$ is twice continuously differentiable in a neighborhood of $m_k(a_0, c)$ and its second derivative is uniformly bounded; similarly, assumption (iii) ensures that $f_{M|A,C}(\cdot)$ is continuous and bounded away from zero in a neighborhood of $m_k(a_0, c)$. Applying a second order Taylor expansion around $m_k(a_0, c)$,

$$\begin{aligned}\mu(m_k(a_0, c) + bu, a_1, c) &= \mu(m_k(a_0, c), a_1, c) + \mu'_m(m_k(a_0, c), a_1, c) bu + \frac{1}{2}\mu''_m(\widetilde{M}_1, a_1, c) b^2 u^2, \\ f_{M|A,C}(m_k(a_0, c) + bu | a_1, c) &= f_{M|A,C}(m_k(a_0, c) | a_1, c) + f'(m_k(a_0, c) | a_1, c) bu + \frac{1}{2}f''(\widetilde{M}_2 | a_1, c) b^2 u^2,\end{aligned}$$

for some $\widetilde{M}_1, \widetilde{M}_2$ between $m_k(a_0, c)$ and $m_k(a_0, c) + bu$. Multiplying these expansions and using that the second derivatives are uniformly bounded yields

$$\begin{aligned}\mu(m_k(a_0, c) + bu, a_1, c) f_{M|A,C}(m_k(a_0, c) + bu | a_1, c) \\ = \mu(m_k(a_0, c), a_1, c) f_{M|A,C}(m_k(a_0, c) | a_1, c) \\ + \{\mu'_m(m_k(a_0, c), a_1, c) f_{M|A,C}(m_k(a_0, c) | a_1, c) + \mu(m_k(a_0, c), a_1, c) f'(m_k(a_0, c) | a_1, c)\} bu \\ + O(b^2 u^2),\end{aligned}$$

where the $O(b^2 u^2)$ term is uniform in u on the support of $\mathcal{K}$.

Substituting into the expression for N_b and using assumption (ii) on the kernel,

$$\int \mathcal{K}(u) du = 1, \quad \int u \mathcal{K}(u) du = 0, \quad \int u^2 \mathcal{K}(u) du < \infty,$$

we obtain

$$\begin{aligned}N_b &= \int [\mu(m_k(a_0, c), a_1, c) f_{M|A,C}(m_k(a_0, c) | a_1, c) \\ &\quad + \{\mu'_m(m_k(a_0, c), a_1, c) f_{M|A,C}(m_k(a_0, c) | a_1, c) + \mu(m_k(a_0, c), a_1, c) f'(m_k(a_0, c) | a_1, c)\} bu \\ &\quad + O(b^2 u^2)] \mathcal{K}(u) du \\ &= \mu(m_k(a_0, c), a_1, c) f_{M|A,C}(m_k(a_0, c) | a_1, c) + O(b^2),\end{aligned}$$

since the term proportional to bu integrates to zero by symmetry of $\mathcal{K}$, and the $O(b^2 u^2)$ term integrates to $O(b^2)$ because $\int u^2 \mathcal{K}(u) du < \infty$.

A similar argument for D_b gives

$$\begin{aligned}D_b &= \int [f_{M|A,C}(m_k(a_0, c) | a_1, c) + f'(m_k(a_0, c) | a_1, c) bu + O(b^2 u^2)] \mathcal{K}(u) du \\ &= f_{M|A,C}(m_k(a_0, c) | a_1, c) + O(b^2).\end{aligned}$$

Assumption (iii) ensures that $f_{M|A,C}(m_k(a_0, c) | a_1, c)$ is bounded away from zero, so for b small enough D_b is also bounded away from zero and we can write

$$\begin{aligned}\mu_{b,k}(a_1, c) &= \frac{N_b}{D_b} = \frac{\mu(m_k(a_0, c), a_1, c) f_{M|A,C}(m_k(a_0, c) | a_1, c) + O(b^2)}{f_{M|A,C}(m_k(a_0, c) | a_1, c) + O(b^2)} \\ &= \mu(m_k(a_0, c), a_1, c) + O(b^2),\end{aligned}$$

where the $O(b^2)$ term is uniform in k and c under the stated uniform boundedness conditions.

Therefore, for each k ,

$$\mu_{b,k}(a_1, c) - \mu(m_k(a_0, c), a_1, c) = O(b^2).$$

Plugging this into the expression for $\widetilde{\Delta}_{h,b}^s(Q)(c)$ and using that $\sum_{k=1}^K g_k(a_0, c) = 1$ yields

$$\begin{aligned}|\widetilde{\Delta}_{h,b}^s(Q)(c)| &\leq \sum_{k=1}^K |\mu_{b,k}(a_1, c) - \mu(m_k(a_0, c), a_1, c)| g_k(a_0, c) \\ &\leq C^* b^2 \sum_{k=1}^K g_k(a_0, c) = C^* b^2,\end{aligned}$$

for some constant $C^* < \infty$ not depending on b . Hence $\tilde{\Delta}_{h,b}^s(Q)(c) = O(b^2)$ as $b \rightarrow 0$, as claimed.

Furthermore, we can write

$$\begin{aligned}\tilde{\Delta}_{h,b}(Q)(c) &:= \tilde{\theta}_{h,b}(Q)(c) - \theta(Q)(c) \\ &= \underbrace{\tilde{\theta}_h(Q)(c) - \theta(Q)(c)}_{\text{coarsening error}} + \underbrace{\tilde{\theta}_{h,b}(Q)(c) - \tilde{\theta}_h(Q)(c)}_{\text{smoothing error}} \\ &= O(w_{\max,K}^2 + b^2).\end{aligned}$$

C.6 THEOREM 5.2

We derive the EIF for the smoothed mediation functional

$$\tilde{\psi}_{h,b}(Q) = \mathbb{E} \left[\sum_{k=1}^K \mu_{b,k}(a_1, C) g_k(a_0, C) \right], \quad (\text{S20})$$

where

$$\mu_{b,k}(a_1, c) = \mathbb{E}[Y \omega_{b,k}(M | a_1, c) | A = a_1, C = c], \quad g_k(a_0, c) = p(\tilde{M} = k | A = a_0, C = c). \quad (\text{S21})$$

Recall that

$$\omega_{b,k}(m | a_1, c) = \frac{\mathcal{K}_b(m - m_k(a_0, c))}{\mathbb{E}(\mathcal{K}_b(M - m_k(a_0, c)) | A = a_1, C = c)}, \quad \mathcal{K}_b(u) = b^{-1} \mathcal{K}(u/b). \quad (\text{S22})$$

The functional $\tilde{\psi}_{h,b}(Q)$ depends on Q in two distinct ways: (i) through the conditional laws defining $g_k(a_0, c)$ and $\mu_{b,k}(a_1, c)$, (ii) through the nuisance center $m_k(a_0, c)$ that appears inside the weight $\omega_{b,k}$.

Along a regular parametric submodel Q_ε with score S , the pathwise derivative decomposes into two contributions,

$$\left. \frac{d}{d\varepsilon} \tilde{\psi}_{h,b}(Q_\varepsilon) \right|_{\varepsilon=0} = \underbrace{\left. \frac{d}{d\varepsilon} \tilde{\psi}_{h,b}(Q_\varepsilon; \{m_k\}) \right|_{\varepsilon=0}}_{\text{vary } Q \text{ holding } \{m_k\} \text{ fixed}} + \underbrace{\sum_{k=1}^K \left. \frac{\partial \tilde{\psi}_{h,b}}{\partial m_k} \right|_Q \left. \frac{d}{d\varepsilon} m_k(Q_\varepsilon) \right|_{\varepsilon=0}}_{\text{chain rule through } m_k}.$$

We compute these two contributions separately.

Part (I): Derivative with respect to the conditional laws (holding m_k fixed). In this first step we treat the function $m \mapsto \omega_{b,k}(m | a_1, c)$ as fixed, so that $\mu_{b,k}(a_1, c) = \mathbb{E}\{Y \omega_{b,k}(M | a_1, c) | A = a_1, C = c\}$ is a standard regression functional; the dependence of the normalization in (S22) on the conditional law is accounted for when we later add the chain-rule correction through m_k .

Let $Z_k \equiv Z_k(O) := Y \omega_{b,k}(M | a_1, C)$. Then $\mu_{b,k}(a_1, c) = \mathbb{E}(Z_k | A = a_1, C = c)$.

For fixed $\omega_{b,k}$, the EIF for the regression functional $c \mapsto \mathbb{E}(Z_k | A = a_1, C = c)$ is

$$\begin{aligned}D_{\mu_{b,k}}^{\text{fix}}(O) &= \frac{\mathbb{I}(A = a_1)}{\pi(a_1 | C)} \{Z_k - \mu_{b,k}(a_1, C)\} \\ &= \frac{\mathbb{I}(A = a_1)}{\pi(a_1 | C)} \{Y \omega_{b,k}(M | a_1, C) - \mu_{b,k}(a_1, C)\}.\end{aligned} \quad (\text{S23})$$

This corresponds to the derivative of $\tilde{\psi}_{h,b}(Q)$ with respect to the conditional distribution argument in the multivariate chain rule.

For $g_k(a_0, c) = \mathbb{E}(\mathbb{I}(\widetilde{M} = k) \mid A = a_0, C = c)$, the EIF is

$$D_{g_k}(O) = \frac{\mathbb{I}(A = a_0)}{\pi(a_0 \mid C)} \left\{ \mathbb{I}(\widetilde{M} = k) - g_k(a_0, C) \right\}. \quad (\text{S24})$$

Since $\widetilde{\theta}_{h,b}(C) = \sum_k g_k(a_0, C) \mu_{b,k}(a_1, C)$ is a product of two regression functionals, its derivative with respect to the conditional laws follows from the usual product rule applied pointwise in C . The EIF for $\widetilde{\psi}_{h,b}(Q) = \mathbb{E}(\widetilde{\theta}_{h,b}(Q)(C))$ under fixed $\omega_{b,k}$ is

$$\widetilde{\phi}_{h,b}^{\text{fixed}}(Q) = \sum_{k=1}^K g_k(a_0, C) D_{\mu_{b,k}}^{\text{fix}}(O) + \sum_{k=1}^K \mu_{b,k}(a_1, C) D_{g_k}(O) + \left\{ \widetilde{\theta}_{h,b}(Q)(C) - \widetilde{\psi}_{h,b}(Q) \right\}. \quad (\text{S25})$$

Substituting (S23) and (S24) yields the explicit displayed formula.

Part (II): Derivative with respect to the nuisance centers $m_k(a_0, c)$. Now consider the actual weight in (S22), which depends on the unknown center

$$m_k(a_0, c) = \mathbb{E}(M \mid A = a_0, C = c, \widetilde{M} = k)$$

and on the normalizing denominator

$$\mathbb{E} \left\{ \mathcal{K}_b(M - m_k(a_0, c)) \mid A = a_1, C = c \right\}.$$

This induces an additional EIF component through the pathwise derivative of $\mu_{b,k}(a_1, c)$ with respect to the center $m_k(a_0, c)$.

The EIF for the conditional mean $m_k(a_0, C) = \mathbb{E}(M \mid A = a_0, C, \widetilde{M} = k)$ is

$$D_{m_k}(O) = \frac{\mathbb{I}(A = a_0) \mathbb{I}(\widetilde{M} = k)}{\pi(a_0 \mid C) g_k(a_0, C)} \left\{ M - m_k(a_0, C) \right\}. \quad (\text{S26})$$

To compute the second term in the multivariate chain rule, we differentiate $\mu_{b,k}(a_1, c)$ with respect to its second argument $m_k(a_0, c)$ while holding the conditional law of $(Y, M) \mid (A = a_1, C = c)$ fixed.

Writing $\mu_{b,k}(a_1, c) = N_k(c)/D_k(c)$ with

$$N_k(c) = \mathbb{E} \left\{ Y \mathcal{K}_b(M - m_k(a_0, c)) \mid A = a_1, C = c \right\}, \quad D_k(c) = \mathbb{E} \left\{ \mathcal{K}_b(M - m_k(a_0, c)) \mid A = a_1, C = c \right\},$$

and applying the quotient rule yields (assuming the usual regularity conditions that justify differentiation under the conditional expectation)

$$\alpha_k(c) := \frac{\partial}{\partial m} \mu_{b,k}(a_1, c) = - \frac{\mathbb{E} \left[\{Y - \mu_{b,k}(a_1, c)\} \mathcal{K}_b'(M - m_k(a_0, c)) \mid A = a_1, C = c \right]}{\mathbb{E} \left[\mathcal{K}_b(M - m_k(a_0, c)) \mid A = a_1, C = c \right]}. \quad (\text{S27})$$

By the multivariate chain rule, the contribution of the nuisance m_k to the EIF is

$$\widetilde{\phi}_{h,b}^{\omega}(Q)(O) = \sum_{k=1}^K g_k(a_0, C) \alpha_k(C) D_{m_k}(O). \quad (\text{S28})$$

Substituting (S26) into (S28) simplifies this term to

$$\widetilde{\phi}_{h,b}^{\omega}(Q)(O) = \sum_{k=1}^K \alpha_k(C) \frac{\mathbb{I}(A = a_0) \mathbb{I}(\widetilde{M} = k)}{\pi(a_0 \mid C)} \left\{ M - m_k(a_0, C) \right\}. \quad (\text{S29})$$

Combining the derivative with respect to the conditional laws and the derivative with respect to the nuisance centers yields the full efficient influence function

$$\widetilde{\phi}_{h,b}(Q)(O) = \widetilde{\phi}_{h,b}^{\text{fixed}}(Q)(O) + \widetilde{\phi}_{h,b}^{\omega}(Q)(O), \quad (\text{S30})$$

where $\widetilde{\phi}_{h,b}^{\text{fixed}}(Q)(O)$ is given in (S25) and $\widetilde{\phi}_{h,b}^{\omega}(Q)(O)$ is given in (S29).

Table S2: Misspecification scenarios for nuisance functions in Simulation #2. A check mark (✓) indicates correct specification, while a cross mark (✗) denotes model misspecification.

Function	Condition 1	Condition 2	Condition 3	Correct	False
m_k	✓	✓	✓	✓	✗
μ	✗	✓	✓	✓	✗
μ_k	✗	✓	✓	✓	✗
g	✓	✗	✓	✓	✗
g_k	✓	✗	✓	✓	✗
π	✓	✓	✗	✓	✗

D DETAILS IN SIMULATIONS

D.1 MISSPECIFICATION SCENARIOS

In this section, we present a detailed explanation of the misspecification scenarios implemented in Simulation #2. For each of the estimators, the corresponding collection of nuisance functions Q is given as follows: (i) $\psi_h(\hat{Q})$: $\{\mu_k, g_k\}$; (ii) $\tilde{\psi}_h(\hat{Q})$: $\{\mu, m_k, g_k\}$; (iii) $\psi_h^+(\hat{Q})$: $\{\mu_k, g_k, \pi\}$; (iv) $\tilde{\psi}_{h_1}^+(\hat{Q})$: $\{\mu, m_k, g_k, \pi\}$; (v) $\tilde{\psi}_{h_2}^+(\hat{Q})$: $\{\mu, m_k, g_k, g, \pi\}$.

Table S2 presents the specification and misspecification configurations of the nuisance functions. The plug-in estimator $\psi_h(\hat{Q})$ coincides with the specification under Condition 3 as well as the fully correctly specified setting; the same holds for $\tilde{\psi}_h(\hat{Q})$. Under the DGP described in Section 6, correctly specified nuisance functions can be consistently estimated using GLMs. Misspecification is introduced by constructing models with nonlinear and irrelevant functional forms, including exponential, cosine, inverse, and higher-order polynomial terms.

D.2 THE PERFORMANCE IN SMALL SAMPLE SIZES

Figure S1 summarizes absolute bias, variance, MSE, and coverage of 95% CIs as functions of K , and Figure S2 presents results under correct specification and a range of misspecification scenarios for nuisance models in a smaller sample size ($n = 500$). The results are consistent with what we find in Simulation 2, where the debiased plug-in estimator $\tilde{\psi}_h(\hat{Q})$ shows substantial improvement compared with the coarsened estimator $\psi_h(\hat{Q})$. The one-step estimator $\psi_h^+(\hat{Q})$ improves bias and coverage relative to $\psi_h(\hat{Q})$, but does not eliminate discretization bias. The first derivative estimator $\tilde{\psi}_{h_1}^+(\hat{Q})$ behaves similarly and does not meaningfully reduce discretization bias. In contrast, $\tilde{\psi}_{h_2}^+(\hat{Q})$ achieves low bias and near-nominal coverage across scenarios. Thus, this simulation confirms the strong performance of our proposed debiased plug-in estimator even in small samples and supports the conclusions drawn in the main text.

D.3 COMPARISON OF DIFFERENT DISCRETIZATION SCHEMES

This section provides additional results for discretization schemes other than the equal-frequency discretization used in the main simulation analysis.

We first examine the population-level coarsening error under equal-width discretization using the same large simulated sample as in Simulation #1. The observed maximum and minimum of M are 6.66 and -5.23 , respectively, giving an empirical range of 11.89 and an equal-width bin length of $11.89/K$. For $K = 2$, this yields the bins $\mathcal{B}_1 = [-5.23, 0.72]$ and $\mathcal{B}_2 = (0.72, 6.66]$. For $K = 6$, the bins $\mathcal{B}_1, \dots, \mathcal{B}_6$ are $[-5.23, -3.25]$, $(-3.25, -1.27]$, $(-1.27, 0.72]$, $(0.72, 2.70]$, $(2.70, 4.68]$, and $(4.68, 6.66]$. Figure S3a plots the within-bin shifts $m_k(a_1, c) - m_k(a_0, c)$ for these two discretization levels. Increasing K reduces these shifts across bins and covariate strata, thereby attenuating the dominant first-order component of the discretization bias. Figure S3b illustrates that, at $C = 0$, with finer discretization, the covariance magnitude drops substantially. These patterns are consistent with the main simulation results.

We then mainly focus on finite-sample estimator performance. Under the same DGP as in Section 6, we compare the coarsened plug-in estimator $\psi_h(\hat{Q})$ with the proposed debiased plug-in estimator $\tilde{\psi}_h(\hat{Q})$ using equal-width and predefined-width discretizations, and discuss the resulting patterns in relation to the equal-frequency results.

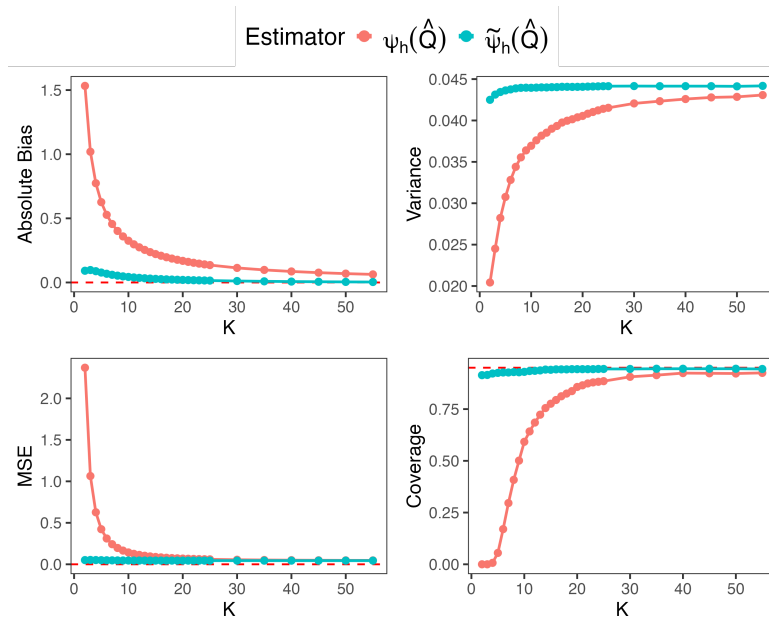


Figure S1: The performance of the coarsened plug-in estimator $\psi_h(\hat{Q})$ and the debiased estimator $\tilde{\psi}_h(\hat{Q})$ as functions of the number of mediator bins K for sample size $n = 500$.

Equal-frequency discretization. The cut points are computed from the observed quantiles of M in each simulated dataset, so each bin contains approximately the same number of observations. This rule makes the empirical bin probabilities stable by construction and makes the interior cut points converge to fixed population quantiles as n increases. Table S3 illustrates the stability of the equal-frequency interior cut points. Consequently, for fixed K , the coarsening bias remains nearly unchanged across the sample sizes in Figure 3. As K increases, the bins become narrower and the coarsened functional becomes more local. The within-bin differences in conditional mediator means, $m_k(a_1, c) - m_k(a_0, c)$, therefore tend to shrink, reducing the leading coarsening bias in $\psi_h(\hat{Q})$. The proposed debiased estimator further removes the first-order bias within each bin, leading to substantial bias reduction.

The variance plots in Figure 4 exhibit the usual bias-variance trade-off induced by finer discretization. As K increases, bins contain fewer observations, so bin-specific quantities such as $\hat{g}_k(a, c)$, $\hat{m}_k(a, c)$, and $\hat{\mu}_k(a, c)$ are estimated with greater variability, leading to increased variance for both estimators. The debiased estimator $\tilde{\psi}_h(\hat{Q})$ generally exhibits slightly larger variance for small K because it relies on a more localized representation of the outcome regression, evaluating $\hat{\mu}$ at the within-bin mean $\hat{m}_k(a_0, c)$ rather than averaging outcomes over the entire bin. This localization removes the leading coarsening bias but provides less smoothing and therefore can increase variability. As K increases, the bins become narrower and $\mu_k(a_1, c) \approx \mu(m_k(a_0, c), a_1, c)$, so the two estimands become increasingly similar and the variance gap correspondingly disappears.

Equal-width discretization. The cut points are computed from the observed minimum and maximum (the observed range) of M in each simulated dataset. Figure S4a shows that, for fixed K except $K = 2$, the bias of the coarsened estimator $\psi_h(\hat{Q})$ can increase with n . This behavior is driven by the unbounded support of the mediator. As the sample size grows, larger samples are more likely to contain extreme tail observations, so the empirical range of M expands. Consequently, for fixed K , the equal-width bin length also increases. As shown in Table S4, the average empirical range increases from 8.11 at $n = 500$ to 10.78 at $n = 50,000$. Because M is unimodally distributed, as illustrated in Figure S6a, this range expansion affects central and tail bins differently. Wider central bins can cover a larger high-density region of M and therefore contain a larger proportion of observations, whereas tail bins will be more sparse and may shift farther into the tails. Table S4 shows that the largest bin proportion among all bins tends to increase with n , especially for small K ; this effect becomes weaker as K increases. The case $K = 2$ is an exception because there is only one interior cut point. Since the distribution of M is only mildly skewed, with skewness -0.29 , the split is relatively stable across n and behaves more similarly to a coarse quantile-based discretization.

To clarify how wider empirical bins affect the coarsening error, we examine the corresponding population-level DGP quantities. As discussed in the main text, the first-order coarsening term in (8) is driven by within-bin differences in

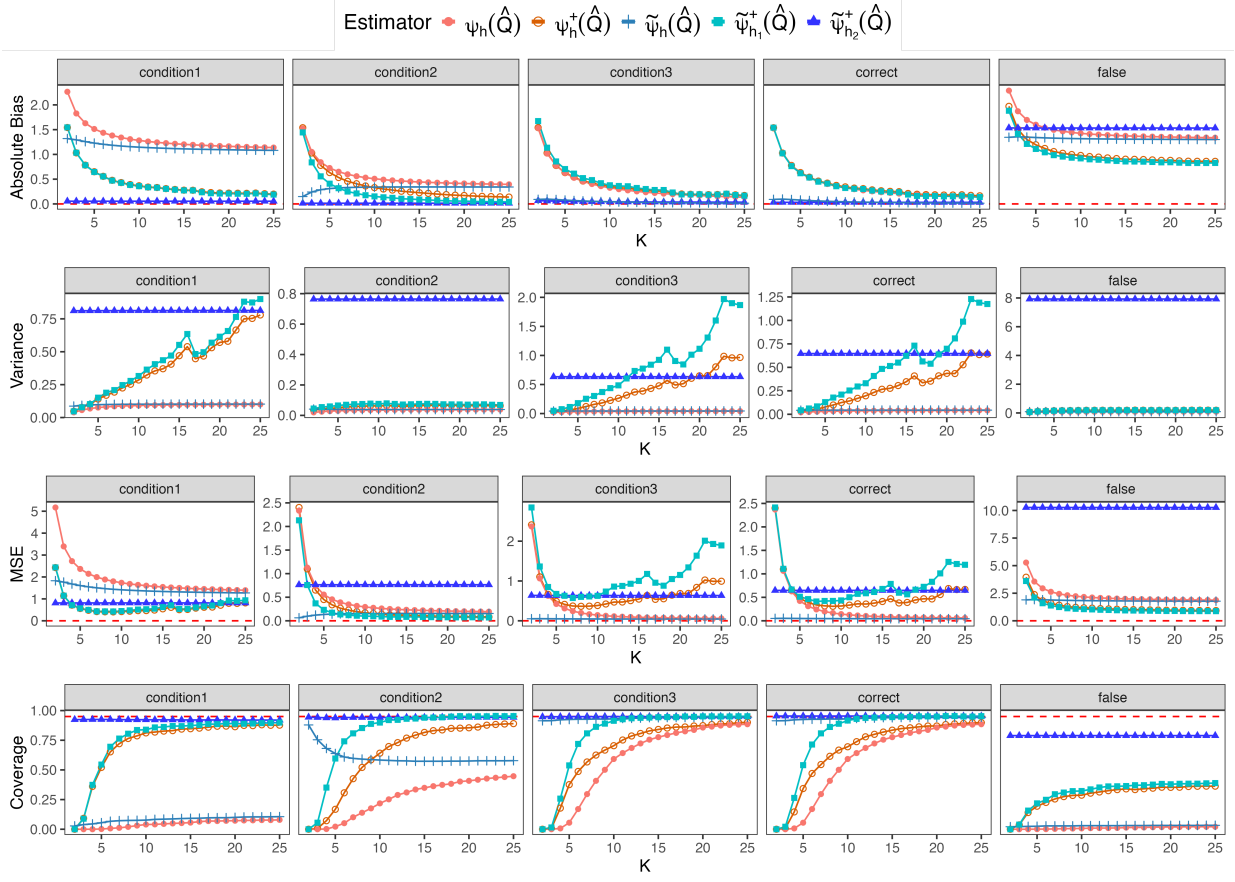


Figure S2: Comparison of plug-in and one-step estimators under nuisance-model misspecification for $n = 500$. “Correct” indicates all nuisance models are correctly specified, whereas “False” indicates all nuisance models are misspecified. Conditions 1–3 correspond to misspecification of $\{\mu, \mu_k\}$, $\{g, g_k\}$, and π , respectively. Performance is summarized in terms of bias, variance, MSE, and 95% confidence interval coverage.

conditional mediator means. In our DGP, the conditional mean of M given $A = a_1$ and $C = c$ is uniformly larger than the conditional mean of M given $A = a_0$ and $C = c$ across the relevant values of c . Therefore, the qualitative pattern is similar across covariate strata, and we use $C = 0$ as an illustrative example. Figure S6b shows that, when the midpoint of a bin is fixed, a bin that contains a larger proportion of the mediator distribution, equivalently a wider bin under this construction, tends to exhibit a larger within-bin difference in conditional mediator means. This increase is more pronounced when the bin midpoint is close to the median. For example, when $K = 3$, the midpoint of the central bin is located near the midpoint of the empirical range. As the central-bin proportion increases from 60.0% to 74.5%, the contrast $m_k(a_1, c) - m_k(a_0, c)$ also increases. By contrast, the tail bins contain less probability mass and move farther into the tails, with their midpoints shifting toward the extremes. As a result, their contributions to the within-bin mean difference are downweighted. Since the curves in Figure S6b are steeper in the central region, changes in the central bin dominate the overall error across bins. Moreover, the leading bias is not determined by $m_k(a_1, c) - m_k(a_0, c)$ alone; it also depends on the derivative of the outcome regression and the bin probability. We therefore examine $\Delta_{h,k}^*(Q)(c) = \mu'_m\{m_k(a_1, c), a_1, c\}\{m_k(a_1, c) - m_k(a_0, c)\}g_k(a_0, c)$, where $\Delta_h^*(Q)(c) = \sum_{k=1}^K \Delta_{h,k}^*(Q)(c)$ and $\Delta_h^*(Q)(c) \approx \Delta_h(Q)(c)$. The diagnostic in Figure S6c confirms that the first-order coarsening component is mainly influenced by the central bins. It also shows that, as K increases, changes in bin proportions become smaller, and the corresponding changes in $\Delta_h^*(Q)(c)$ are attenuated. For example, when $K = 10$, $\Delta_h^*(Q)(c)$ changes only modestly as the central-bin proportion increases from 22.4% to 28.3%. This behavior of the population-level DGP quantities explains the increasing bias trend of the coarsened estimator $\psi_h(\hat{Q})$ shown in Figure S4a.

Figure S5a further shows that the variance of $\psi_h(\hat{Q})$ under equal-width discretization can be nonmonotone in K . This behavior can be understood by decomposing the variability into two components: the usual sampling variability, as in equal-frequency discretization, and the additional variability induced by the random partition. Under empirical equal-width

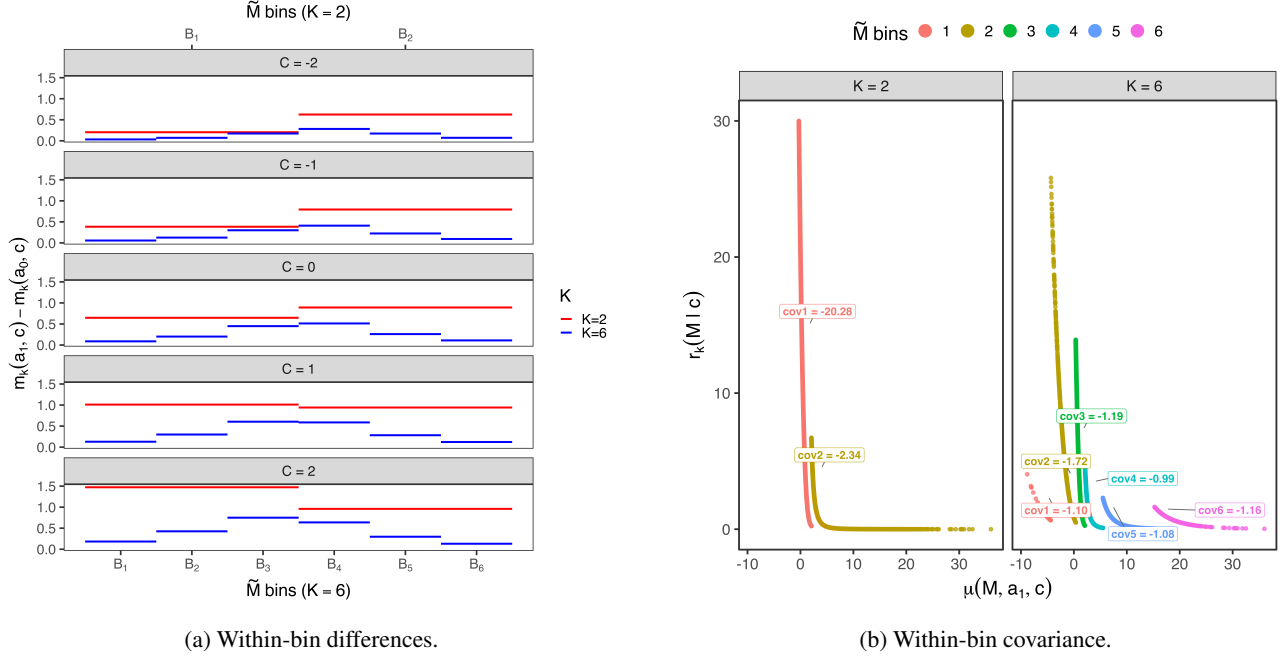


Figure S3: Theoretical quantities approximation in large sample size. (a) Within-bin differences $m_k(a_1, c) - m_k(a_0, c)$ under $K = 2$ (red, upper axis) and $K = 6$ (blue, lower axis) equal-width discretizations. Each segment represents one mediator bin; panels correspond to values of C , and (b) Within-bin covariance between $\mu(M, a_1, C)$ and $r_k(M | C)$ at $C = 0$.

discretization, the partition is itself random because small changes in the sample range can move the interior cut points. These changes alter both the bin probabilities and the within-bin contrasts that enter the coarsened target. For larger K , the bins become more local and the sensitivity of the coarsened functional to any single boundary is reduced. In contrast, the debiased estimator $\tilde{\psi}_h(\hat{Q})$ is substantially less affected by this random-partition behavior because it removes the leading first-order coarsening component within each bin. As K increases, the variance pattern of $\psi_h(\hat{Q})$ becomes closer to the pattern observed under equal-frequency, where finer partitions reduce smoothing and tend to increase variance more gradually.

Predefined-width discretization. The cut points are fixed across simulated datasets. For each K , we partition the interval $[-1.5, 3.3]$ into K subintervals of equal length and set the outer boundary cut points to the observed minimum (ℓ) and maximum values (u), so that observations outside the fixed interior interval are included in the first and last bins. For example, when $K = 3$, the bins are $[\ell, 0.1]$, $(0.1, 1.7]$, and $(1.7, u]$. Thus, for a given K , the same cut points are used in every Monte Carlo replication.

Because the interior bin widths do not depend on the empirical range, the bias of $\psi_h(\hat{Q})$ remains stable across n for fixed K , as shown in Figure S4b. This pattern is similar to that observed under equal-frequency discretization. However, although the bias decreases as K increases, it converges to a nonzero value. This occurs because increasing K refines the partition only within the fixed interval $[-1.5, 3.3]$, while the two tail bins always contain the regions ≤ -1.5 and ≥ 3.3 . Once the central-region discretization bias is reduced, the remaining approximation error may be dominated by these tail bins. This explains why residual tail-induced bias can persist under predefined-width discretization even for large K .

Choice of discretization scheme and discretization levels. These results highlight complementary practical considerations for choosing a discretization scheme. Equal-frequency discretization balances empirical bin masses and is often stable in finite samples, since the cut points converge to population quantiles. Empirical equal-width discretization is closely aligned with the bounded-support statement in the main paper, but it can be sensitive to sample extremes when the mediator has unbounded support, because the empirical range may expand with the sample size. Predefined-width discretization is useful when categories are fixed by scientific or clinical convention, as in settings where thresholds are externally defined (e.g. BMI) and should not vary across studies. However, this approach lacks the flexibility to adapt the choice of K to a specific dataset. Moreover, if the mediator has non-negligible probability mass within coarse fixed intervals, such bins may induce residual coarsening bias. In such settings, the proposed debiased estimator $\tilde{\psi}_h(\hat{Q})$ is particularly useful, since it

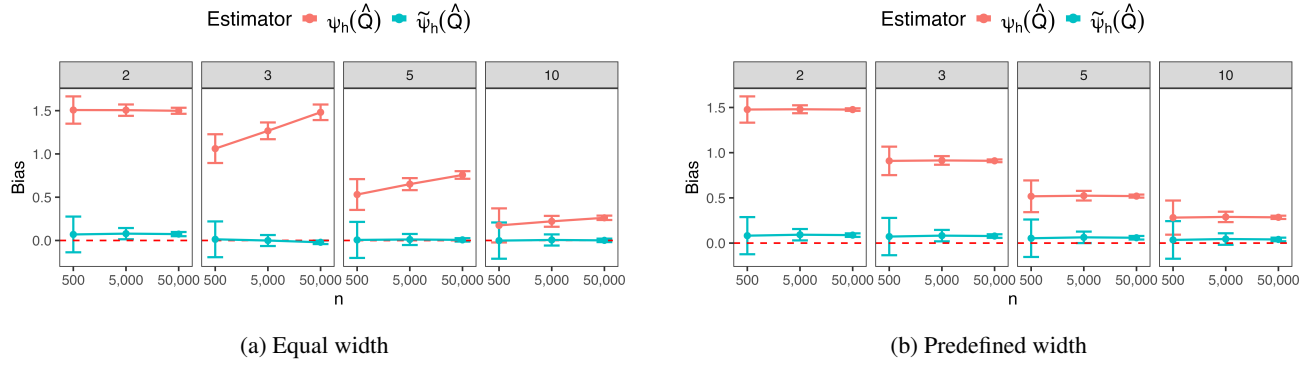


Figure S4: Bias trends of the coarsened plug-in estimator $\psi_h(\hat{Q})$ and the debiased plug-in estimator $\tilde{\psi}_h(\hat{Q})$ as the sample size n increases under empirical equal-width and predefined-width discretizations.

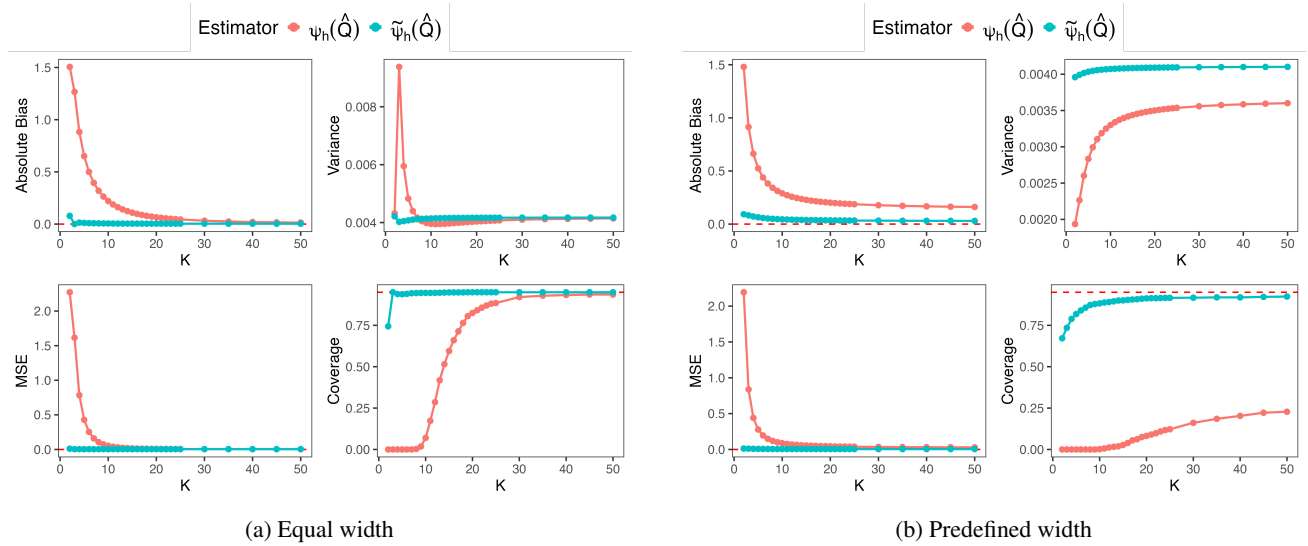


Figure S5: Performance of the coarsened plug-in estimator $\psi_h(\hat{Q})$ and the debiased estimator $\tilde{\psi}_h(\hat{Q})$ as functions of the number of mediator bins K for sample size $n = 5,000$ under different discretization schemes.

reduces sensitivity to the chosen discretization scheme and substantially reduces bias for small K , while preserving the computational simplicity of discretization.

In practice, in addition to the choice of discretization scheme, the choice of the number of discretization levels K is also an important consideration. From a theoretical perspective, increasing K reduces approximation bias, but it may increase estimation variance and computational burden, particularly when some bins contain few observations. An implication of our results is that the proposed debiased estimator substantially reduces bias relative to the naive coarsened estimator even for small K . When scientifically meaningful categories are available, such as standard clinical thresholds, we recommend using the proposed debiased estimator with those bins. When no natural discretization is available, a practical strategy is to start with a small K and gradually increase it to assess stability. If the estimates remain stable for larger values of K , one may select the smallest stable K for simplicity.

E A REAL DATA APPLICATION

We apply our proposed estimation framework to data from the B_PROUD study, an observational investigation assessing the impact of Mobile Stroke Unit (MSU) dispatch on post-stroke outcomes in Berlin [Ebinger et al., 2017] conducted between February 2017 and May 2019. It can be used to evaluate the effect of additional MSU care on 3-month functional outcomes among patients for whom an MSU was received.

Table S3: Summary of interior break points under equal-frequency discretization for $K = 2$, $K = 3$, and $K = 5$ across Monte Carlo replications. Values are reported as mean $\pm$ standard deviation. The boundary cut points are fixed at $-\infty$ (0%) and ∞ (100%).

n	$K = 2$	$K = 3$		$K = 5$			
	50.0%	33.3%	66.7%	20.0%	40.0%	60.0%	80.0%
500	1.21 $\pm$ 0.08	0.53 $\pm$ 0.10	1.81 $\pm$ 0.08	-0.16 $\pm$ 0.11	0.82 $\pm$ 0.09	1.57 $\pm$ 0.08	2.33 $\pm$ 0.08
5,000	1.21 $\pm$ 0.03	0.53 $\pm$ 0.03	1.81 $\pm$ 0.03	-0.16 $\pm$ 0.03	0.82 $\pm$ 0.03	1.57 $\pm$ 0.03	2.34 $\pm$ 0.02
50,000	1.21 $\pm$ 0.01	0.53 $\pm$ 0.01	1.81 $\pm$ 0.01	-0.17 $\pm$ 0.01	0.82 $\pm$ 0.01	1.57 $\pm$ 0.01	2.34 $\pm$ 0.01

Table S4: Summary of the empirical range, minimum, maximum, and the largest bin proportions under equal-width discretization across Monte Carlo replications. Values are reported as mean $\pm$ standard deviation.

n	Range	Min	Max	Maximum proportion in bins (%)				
				$K = 2$	$K = 3$	$K = 5$	$K = 10$	$K = 20$
500	8.11 $\pm$ 0.57	-3.30 $\pm$ 0.44	4.81 $\pm$ 0.39	61.6 $\pm$ 6.4	60.0 $\pm$ 4.9	39.6 $\pm$ 3.4	22.4 $\pm$ 2.0	12.3 $\pm$ 1.2
5,000	9.56 $\pm$ 0.51	-4.08 $\pm$ 0.38	5.48 $\pm$ 0.33	62.7 $\pm$ 5.6	68.3 $\pm$ 3.7	44.0 $\pm$ 3.0	25.4 $\pm$ 1.4	13.3 $\pm$ 0.8
50,000	10.78 $\pm$ 0.43	-4.73 $\pm$ 0.32	6.05 $\pm$ 0.28	63.6 $\pm$ 4.7	74.5 $\pm$ 3.0	48.4 $\pm$ 2.8	28.3 $\pm$ 1.2	14.8 $\pm$ 0.6

We analyzed 768 patients eligible for reperfusion therapy, of whom 588 (76.6%) received MSU care ($A = 1$) and 180 (23.4%) received conventional emergency services ($A = 0$). The outcome Y is the 3-month modified Rankin Scale (mRS), an ordinal measure ranging from 0 (no symptoms) to 6 (death). In our study, we treat mRS as continuous. The full mediator M is defined as the time from ambulance dispatch to thrombolysis and is set to zero for patients who did not receive thrombolytic therapy. We adjust for two baseline continuous covariates: systolic blood pressure and stroke severity. Missing data are handled using multiple imputation.

This dataset was previously analyzed by Piccininni et al. [2023] employing a front-door approach to estimate the causal effect of MSU dispatch on 3-month mRS using a three-category mediator. Subsequently, Guo et al. [2023] considered both continuous and binary versions of the mediator, exploring different cutoffs and implementing one-step and TMLE estimators of the front-door functional. In this analysis, we compare three estimators of the front-door functional $\gamma(Q)$ for $\mathbb{E}(Y(a_0))$ in (2) with $a_0 = 0$: (i) the coarsened plug-in estimator $\gamma_h(\hat{Q})$, (ii) the debiased plug-in estimator $\tilde{\gamma}_h(\hat{Q})$, and (iii) the plug-in estimator $\gamma_s(\hat{Q})$ based on sequential regression, which avoids explicit estimation of the mediator density by rewriting $\theta(Q)(c)$ as

$$\theta(Q)(c) = \mathbb{E}(\tilde{Y} | a_0, c), \quad \text{where} \quad \tilde{Y} = \mathbb{E}(Y | M, a_1, c). \quad (\text{S31})$$

All nuisance functions were estimated using super learner with candidates including `glm.interaction`, `glmnet`, `random forests` (1000 trees), `xgboost`, and `dbarts`.

Figure S7 presents point estimates along with 95% CIs obtained via bootstrapping. In terms of point estimates, the debiased estimator $\tilde{\gamma}_h(\hat{Q})$ and the sequential estimator $\gamma_s(\hat{Q})$ are closer to each other than either is to the coarsened estimator $\gamma_h(\hat{Q})$. When $K = 2$, the CIs of the three estimators exhibit limited overlap, and $\gamma_h(\hat{Q})$ (2.074; 95% CI: 1.931–2.217) appears somewhat farther from the debiased estimator $\tilde{\gamma}_h(\hat{Q})$ (2.182; 95% CI: 2.015–2.382) and the sequential estimator $\gamma_s(\hat{Q})$ (2.157; 95% CI: 1.984–2.340). For $K = 3, 4$, and 5 , the CIs largely overlap. In particular, the estimate of $\tilde{\gamma}_h(\hat{Q})$ with $K = 5$ (2.144; 95% CI: 2.010–2.340) is nearly identical to the estimate of $\gamma_s(\hat{Q})$, indicating substantial agreement under finer discretization.

The coarsened estimator $\gamma_h(\hat{Q})$ does not perform as poorly as might be anticipated. This pattern is interpretable upon examining the within-bin mediator mean shifts across treatment levels, as illustrated in Figure S8. For $K = 2$, the estimated shifts in the two bins cluster around opposite signs: one predominantly positive and the other predominantly negative. As indicated in (6), the coarsening bias depends on both the mediator shifts within bins and the local derivative structure of $\mu(m, a_1, c)$. When mediator shifts differ in sign across bins, their weighted contributions may partially offset. This cancellation effect provides a plausible explanation for the relatively small empirical deviation of the coarsened estimator compared with the other estimators in this dataset.

However, such offsets depend on specific features of the data structure and cannot always be guaranteed in practice.

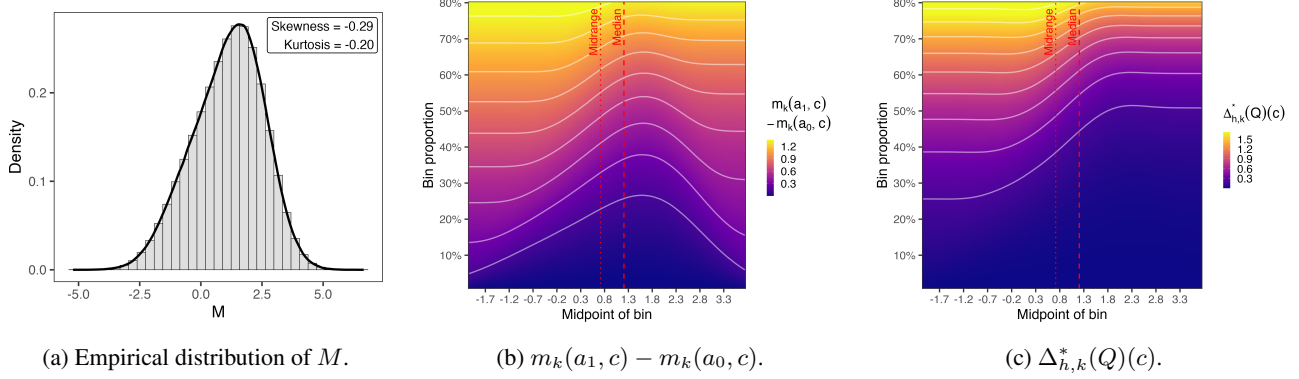


Figure S6: Diagnostic plots for mediator bins (covariate $C=0$). Panel (a) shows the empirical distribution of the mediator M in a large simulated sample. Panel (b) shows the within-bin difference in conditional mediator means, $m_k(a_1, c) - m_k(a_0, c)$. Panel (c) shows the first-order coarsening-bias component, $\Delta_{h,k}^*(Q)(c) = \mu'_m(m_k(a_1, c), a_1, c) \{m_k(a_1, c) - m_k(a_0, c)\} g_k(a_0, c)$. Panels (b) and (c) are evaluated from the theoretical functions: for a bin with a given proportion on the y -axis, the color represents the corresponding value of the target quantity as the bin midpoint varies along the x -axis. The dotted line represents the midpoint of the range and the dashed line represents the median point.

Therefore, although the coarsened plug-in estimator may perform well in certain empirical settings, relying on it without bias correction remains potentially risky. In contrast, the proposed debiased estimator offers a more reliable alternative, mitigating first-order coarsening bias without imposing restrictive assumptions on the mediator–outcome relationship.

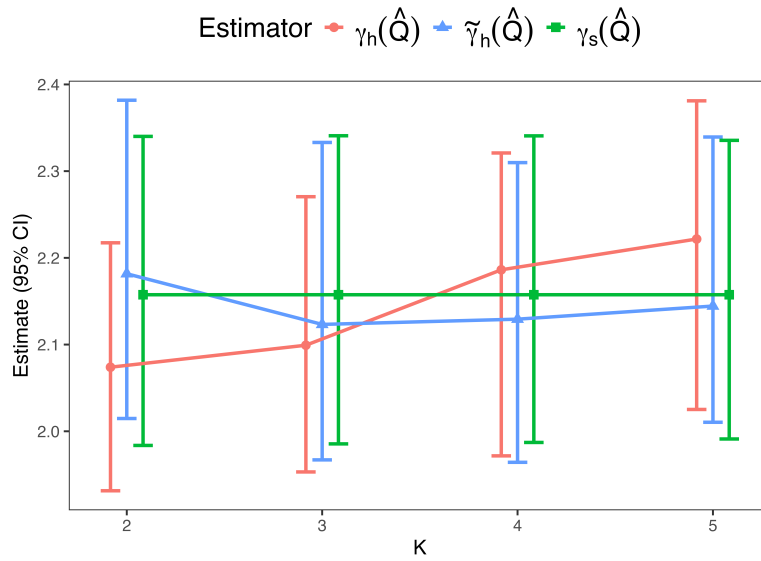


Figure S7: Comparison of plug-in estimators $\gamma_h(\hat{Q})$, $\tilde{\gamma}_h(\hat{Q})$ and $\gamma_s(\hat{Q})$ in B_PROUD study for estimating $\gamma(Q) = \mathbb{E}[Y(a_0)]$

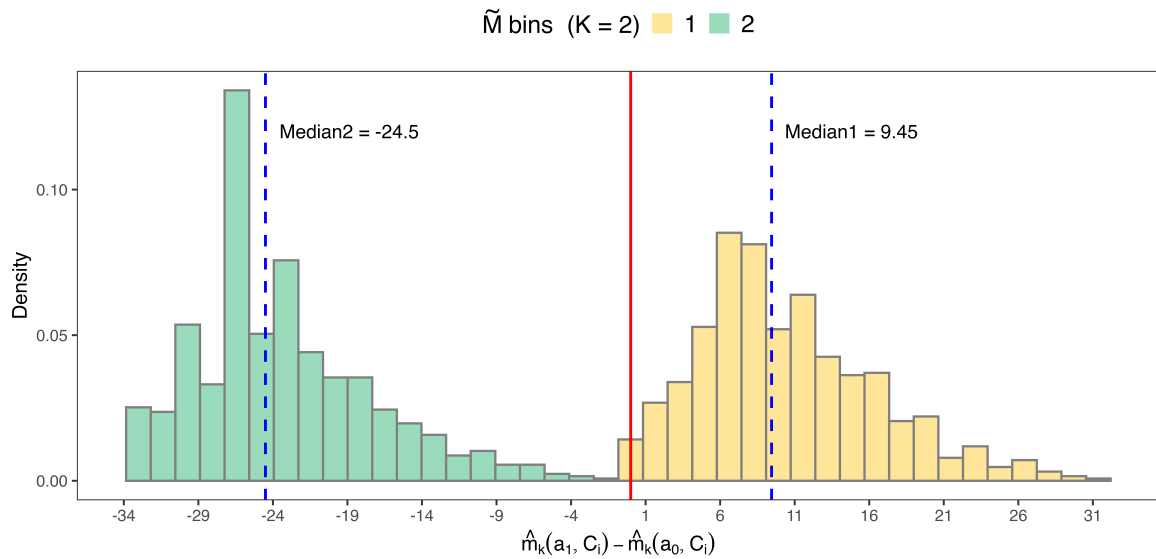


Figure S8: Estimated within-bin differences in conditional mediator means, $\hat{m}_k(a_1, C_i) - \hat{m}_k(a_0, C_i)$, under discretizations with $K = 2$. The red line denotes zero and the blue dashed lines indicate the median of within-bin differences.